LEVERAGING WEB SCRAPING AND AI AGENTS FOR SCAM DETECTION ON SOCIAL MEDIA

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STATEMENT OF CANDIDATE

I, Nicholas Simos, declare that this report, submitted as part of the requirement for the award of Bachelor of Engineering in the School of Engineering, Macquarie University, is entirely my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualification or assessment an any academic institution.

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Date: 8th June 2025

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STATEMENT OF USING GENERATIVE AI AND EDITING TOOLS

I, Nicholas Simos, acknowledge the use of artificial intelligence (AI) tools in the research project and the writing of this thesis. In this work:

- I used ChatGPT v4 to help with prompt engineering for initial project prototype
- I used ChatGPT v4 to help Complete missing fields in BibTeX References
- I used Grammarly to help with checking grammar and clarity.
- I used OpenAI gpt-4o applied in the Thesis A prototype for Scam classification and Scoring
- I used Anthropic Claude-3-7-sonnet-20250219 to filter data collected in Thesis A prototype.

I am aware of the limitations of these tools and am ultimately responsible for the content, research and developing my ideas described in this document.

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ABSTRACT

Scams on social media platforms have become an increasing threat, targeting individuals and organizations around the world with malicious and deceptive tactics for personal gain. This thesis investigates how AI-driven agents and web scraping techniques can be combined to effectively detect and track evolving scam activities. An examination of the literature and a case study regarding the social media platform "Facebook" reveals that current efforts for moderation are reactionary, only able to detect scams following known scam trends after they appear. Additionally, it is identified how artificial intelligence has been integrated into both newly emerging scams and scam detection methods and how both utilize web scraping and automation techniques to operate autonomously. In the approach to address this, a system is proposed to be developed following a multi agent architecture to include multiple autonomous agents for web scraping, crawling, scam classification, and trend analysis that operate as a cohesive unit to detect and identify scam trends in real time.

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Chapter 1

Introduction

Social media platforms and their provided communication systems have rapidly become an indispensable aspect of life for billions of people around the world, changing the way many communicate and share information. However, it is this instant global connection and reach that has fostered the perfect environment for exploitation of vulnerable users through increasingly sophisticated scam activities. Statistics from the National Anti-Scam Center show that Australians lost $2.03 billion dollars (AUD) in 2024 with 74% of that coming from investment, romance, payment redirection, remote access and phishing scams alone [11]. These types of scams and many more used by malicious entities on social media platforms have grown ever more sophisticated in recent years with the introduction of Artificial Intelligence (AI) and even with a %17 reduction in scam reports, Australians are still forecast to lose more than what was recorded in 2024 [17].

Social media and communication platforms employ a combination of human and Artificial Intelligence based moderation techniques in an effort to prevent such events from occurring, but with the ever evolving nature of scam activity on the Internet, these techniques have proven to be ineffective and only provide a reactionary response to commonly known scam trends [31, 36]. Additionally through their communication systems moderation techniques are only employed to counter un-desired human interaction with no regards to the context or content shared. Transparency reports published by Meta (Facebook, Instagram, WhatsApp) and X (Twitter) under the European Union Digital Services Act (EU DSA 2022) further highlight the lack of successful moderation systems and effort to deploy countermeasures through the under-representation of data related to scam content moderation [46, 72]. It is this void in evolving moderation techniques regarding scam, introduction of artificial intelligence generated scam content, and declining trend in scam incident reports that indicate the necessity for advanced and adaptive countermeasures against malicious organizations on social media platforms in order to prevent further harm in the future [11].

Artificial Intelligence has rapidly evolved into an integral part of society with profound impact in nearly all parts of daily life and across nearly all industries around the world.

The beneficial impact of AI has positioned itself as a driving force of innovation in the digital world, healthcare, finance, and many other aspects of our global ecosystem, but it is this positioning and accelerating ascent of reliance that has revealed its ability to be used as a doubled edged sword [40, 55]. Alongside the immense benefits and widespread deployment it has introduced new unforeseen risks and challenges in the realm of cyber-security as it has empowered scammers with sophisticated tools and techniques to elevate scammers already elaborate operations [40]. Parallel to the advancements in artificial intelligence, the technique of large scale Open Source Intelligence (OSINT) gathering through web scraping, the process of automated information extraction from the internet has become a core component of scam operations. Just as AI can have both negative and positive effects, web scraping can also be considered to have similar capabilities. Web Scraping has provided the ability to empower legitimate platform moderation and scam intelligence investigations but it has also provided cyber-criminals with additional tools to gather information about users to fuel their malicious operations [2]. Web scraping data from social media platform profiles and public databases in order to gather information on potential victims is the techniques driving force for their ability to identify potential victims that may be unaware of the potential threats they are exposed to while on Social Media Platforms. Although this is the case the technique of web scraping can also be beneficial in gathering real-time data to be used in counteracting the actions of malicious actors [2].The combination of both artificial intelligence and web scraping allows the complementary strengths of both tools to create an adaptive real time scam detection system. The work in this thesis is situated to explore how AI driven agents augmented with web scraping techniques can effectively detect and track evolving scam activities on social media and provide a proactive and adaptive solution to malicious activity in order to combat the growing presence of cyber-criminals.

1.1 Project Goal

The goal of this project is to design and develop an autonomous, AI driven multi agent system that incorporates web scraping and browser automation techniques to detect and track evolving scam activities and trends on a social media and communication platforms in real time. The system objective is to search the internet for scam related content/groups/chats on social media platforms or the wider Internet, and gather the data required to produce an evaluation on wether the content is potentially scam related in real-time. Additionally the system will also provide a high level overview on evolving scam trends through the use of the collected data including context, continuous evaluations and confidence scoring.

The system will be comprised of multiple specialized agents each with an individual task to complete, including a web scraper agent, scam analysis agent, and a supervisor agent, collaborating together to gather, process and analyses scam data from social me-

dia environments. The collective work of these agents will result in an adaptive data collection solution that can be used for advanced moderation of social media platforms and groups on their communication systems.In the case of testing and initial deployment the target platforms will be Facebook, WhatsApp, Telegram, TikTok, and LinkedIn. The goal of the web scraper agent will be to collect the relevant scam related data needed for the scam analysis agent to produce an evaluation on the content gathered and then additionally generate a confidence score. The collective data gathered during this process will be analyses to develop real-time scam trend information by the supervisor agent and will be fed back to the agent pipeline to adjust search algorithms and individual scam content analysis results. This Closed-Loop Knowledge Sharing Multi Agent (CLMA) architecture will be evaluated against a traditional Feed-forward Agent Pipeline (FFP) architecture to evaluate the overall performance and adaptability over a set period of time.

Through the automation of searching and analyzing scam related content and groups on social media platforms in real time, the thesis project aims to address the significant gap in the current moderation and detection systems employed by organizations in an effort to gather information on the evolving scam content on the Internet. The resulting system intends to be a proactive solution capable of dynamically adapting to emerging scam patterns and threats through the use of real-time analysis to direct its search and develop a deeper understanding of emerging scam tactics.

1.2 Gantt Chart / Project Timeline

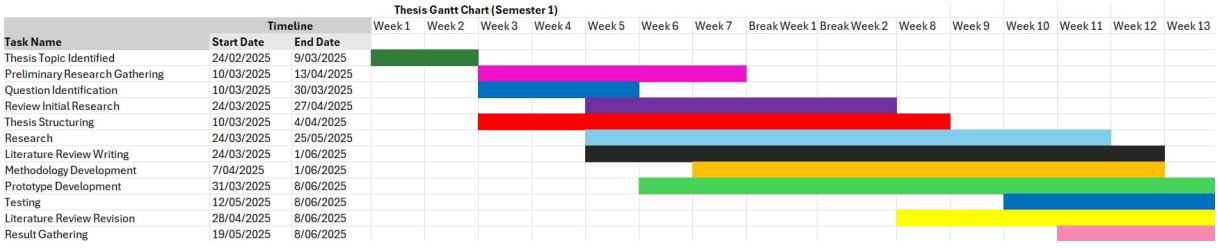


Figure 1.1: Thesis A Gantt Chart - Timeline

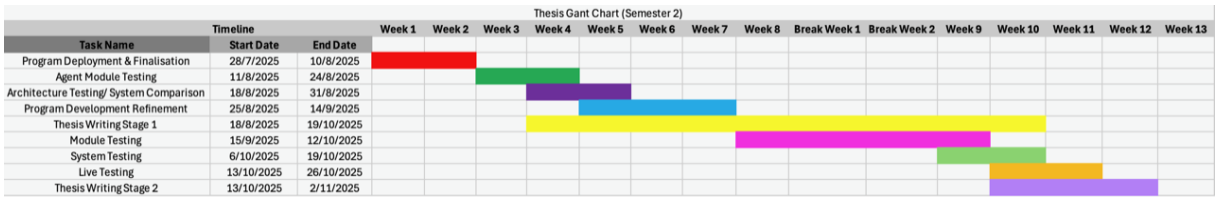


Figure 1.2: Thesis B Gantt Chart - Timeline

1.3 Thesis Questions

The following are the main research questions along with their respective subproblems that will be addressed within this thesis document:

1. How can AI-driven agents and web scraping techniques be combined to effectively detect and track evolving scam activity on social media and communication platforms?
    (a) Can multiple specialized Large Language Model based AI-Agents be combined with system self-reflection loops to detect evolving scam trends using scam content in real-time in a multi-agent system?
    (b) Can we effectively use generative AI to generate insightful responses on social media or communication content in efforts to prevent potential scams?
2. What modern web scraping/crawling strategies are effective when dealing with dynamic, security hardened social media and communication platforms?
    (a) How can modern web scraping tools and techniques be used in conjunction with Large Language Models to collect data across multiple domains?
3. How can Large Language Model powered AI Agents be utilized to identify and classify potential scam content on communication platforms?
    (a) How can we validate and evaluate the effectiveness the the system in its entirety through Large Language Model (LLM) generated context responses?
4. Does a Closed-Loop Knowledge Sharing Multi Agent architecture (CLMA) with an Orchestrator outperform an Open Loop Feed-forward Agent Pipeline (FFP) architecture for real-time scam detection on social media and communication platforms?
5. How can software engineering practices be applied to ensure development of adaptable, fault-tolerant, and robust systems in an automated scam detection and classification Multi Agent Architecture?
    (a) Does the architecture of a system impact these factors and increase system performance?

The following paper is structured as follows, Chapter 2 is a literature review that presents relevant contextual content and a comprehensive critical analysis of published and credible sources around the papers main topic and research questions. Chapter 3 is a high level system overview that briefly describes the systems being developed and compared, the system requirements, and a brief description of the project Agents making up the complete system. Chapter 4 provides in-depth detail about the system architectures and modules as well as all methodology details. This chapter is a detailed analysis of all core features, system operations, and technologies. Chapter 5 outlines the testing methodology used

for the systems being compared and results being collected. Chapter 6 presents all the results as part of the completion of the testing process as well as additional detail about performance and final results. Chapter 7 is an critical analysis of the findings from the results and a discussion regarding the evaluation of these results and how they present answers for the research questions of this thesis paper. Chapter 8 is an outline of future works to be completed as apart of further development and refinement of the system moving forward. Finally Chapter 9 is a conclusion highlighting key points and final throughts regarding the thesis paper

Chapter 2

Literature Review

2.1 Social Media as a Scam Environment

In today's increasingly digital world, online social network platforms such as Facebook, Instagram, X (Twitter), WhatsApp, Telegram, or LinkedIn have become an indispensable part of modern society and have seen significant growth with their ability to provide open nature communication and information sharing to billions of users around the world. However, scam operations have proliferated alongside the same growth of social media platforms with significantly increasing effects, as reported by the Global Anti-Scam Alliance (GASA), with an increase of 1772.73% lost globally from 2020 to 2024 [25, 64]. In contrast to this dramatic increase, GASA also reported that around 70% of victims did not report that they had been scammed to the proper authorities or organizations as its estimated that over 42% of these criminal acts were carried out on social media platform alone [2, 25, 26, 64]. This dramatic disparity between the total financial loss to scams globally and the significantly lower number of reports highlights the substantial gap in effective moderation and preventive systems in place to protect users and demonstrates how social media platforms have become an ever-growing fertile ground for cyber criminals [25, 26, 51].

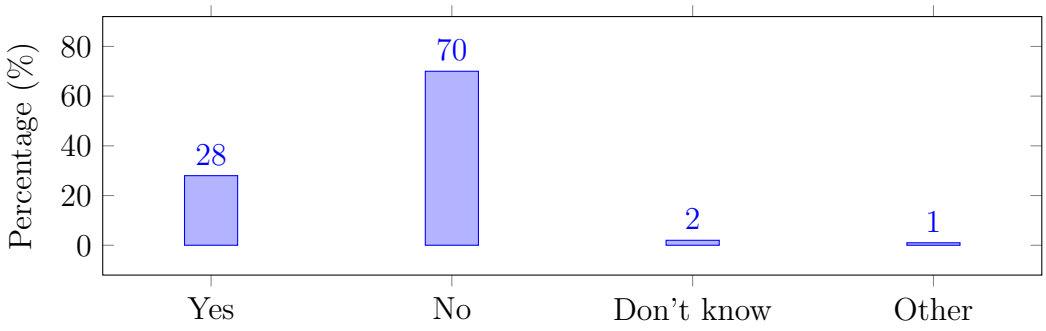


Figure 2.1: Proportion of scam victims who reported incidents to the proper authorities. [25]

Social media platforms allow users to communicate, connect, share content, build an online presence, access information or entertainment, and participate in public discussion as all these components are becoming increasingly easier with the integration of AI-agents to assist platform users in all aspects. Although, it is these key foundational components that are utilized in conjunction with malicious techniques and cyber criminals own adoption of AI to deceive unknowledgeable users. Fundamentally, the virtualization of key societal components is the foundation of platforms such as Facebook, Reddit, Instagram, WhatsApp, and more, but through this virtualization scams are able to flourish as they are obscured by anonymity, deception and the trust users have built with social media platforms. [40]. Additionally, with social media platforms having conjoining communication applications, Figure 2.2 displays data regarding the method of first contact and not the overall number of contact methods employed in the process of conducting the scam.

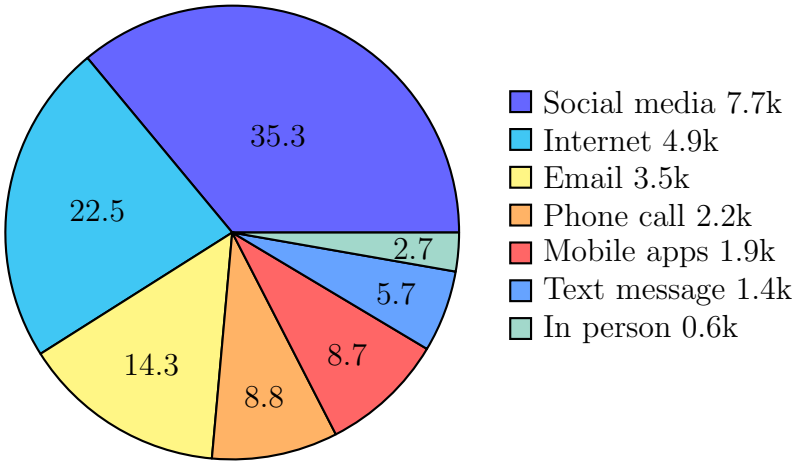


Figure 2.2: Top Contact methods by number of reports with loss [17]

With social media becoming the most common form of contact that leads to financial loss and the alarming rate of scam material being generated on the platforms due to the introduction of Artificial Intelligent Agents (AI-Agents), cyber criminals are capitalizing on the baseless trust many users have with platforms [6, 59]. Additionally, social media platforms market a new level of moderation through the use of artificial intelligence but there is no evidence or literature to support these claims regarding scam activity on social media platforms. Current systems of moderation have been identified to focus on user behavior factors/language (for example, high number of messages in a short period of time, mature language, etc), these systems use a specified baseline behavioral pattern and existing user data to identify outliers which don't fit the platforms expected user behavior. These outliers are categorized based on the difference in their user data to a preset expected user and are removed from the platform if considered to be harmful. While traditional cyber attacks previously exploited technical vulnerabilities, evolving malicious activity has shifted to scams that prey on human psychology and social engineering, knowing there are no systems in place to prevent emerging scams of this type. [36, 46, 72].

2.1.1 Background Review - Typology of Scam Activities on Social Media and Communication Platforms

Social media scams are specifically tailored for their respective platform to utilize their provided capabilities and improve the effectiveness of the specific type of scam. Though this may be the case, cyber criminals still deploy a vast number of different scams in efforts to reach a wider range of potential victims, but for nearly all scams, social media platforms are still identified as the leading initial connection point between the two parties. The literature and statistical data from various governing bodies around the world strongly indicate that this is the case, as the annual transparency reports published by these bodies all communicate that a select number of scams with the greatest economic impact are predominantly engaged via social media platforms [10, 12, 16, 21, 25, 52, 65]. The Australian Government National Anti-Scam Center and CERT New Zealand both highlighted this as within the 2023 to 2025 time period Facebook was identified as the leading platform for scam activity online [11, 18]. This is also supported through data published by the Europol Internet Organized Crime Threat Organization as it displays how social media is a key factor in connecting international cyber criminals with potential scam victims and further highlights that organizations such as Meta (Facebook, WhatsApp) and X (Twitter) have provided little support in counteracting this [23, 71]. Although evidence identifies Meta and X as the primary platforms for cyber criminals and scam activity, the under representation of scam-related data by these organizations' own mandatory transparency reports may paradoxically reinforce this claim, suggesting a potential gap in detection, public accountability, and moderation regarding the true scale of fraudulent activity on the platforms [41, 46, 72].

Within the environment of social networks, cyber criminals employ a diverse range of strategies, combining social engineering tactics and technical subterfuge to enhance the effectiveness of scams and exploit vulnerable users. A survey conducted by Rathore (2017) and colleges classifies social media security threats and scams into three main categories, multimedia content threats, traditional communication technique threats, and social engineering threats, with an overall of 21 different types of scams encapsulated in these classifications [61]. These scams manifest themselves in a variety of forms, with the most commonly identified across reviewed literature being:

Investment / Cryptocurrency Scams

Investment scams are fraudulent schemes with the goal of tricking individuals into handing over funds for stock trades, binary options, foreign exchanges, or Crypto currency offers with promises of high returns at minimal to no risk. Often investment scams are used in conjunction with impersonation scams and employ social engineering techniques with falsified success stories or fraudulent screenshots of potential profits to get the victim to transfer funds or invest in fake investment platforms [5, 24, 49]. The target for scams within this category are often people in financial hardship or from low socio-economic areas

looking for a means to resolve their current situation. Victims in these circumstances are targeted as the promise of high returns compromises their judgment making them more susceptible to impulsivity and offers that require urgent action. In current times cryptocurrency scams have become the leading investment scam method on social media platforms with many cyber criminals using additional scam techniques such as fraudulent advertisement or impersonation to funnel victims off the platform and onto websites or investment platforms where they will either take the victims money or lead them to invest into "pump and dump" schemes [47, 63]. The main reason for this shift in investment scam focus by cyber criminals is because modern currency trading platforms are unregulated, open to corporate operation with little regulation, and allow them to steal victims money through self-generated market manipulation [47,63]. Zauwiyah (2024) identifies this segment of investment scams as the next evolution in the world of scam and further review of literature material support these facts as cryptocurrency gains further foundational footing in today's digital society [5, 47, 63].

Phishing Scams

Phishing scams are attempts to deceive users into giving sensitive information through the impersonation of tested entities. Often phishing scams are used in conjunction with impersonation scams where the scammers pose as employees to lure victims into clicking malicious links in hopes of either getting victims to download malicious programs or divulge classified information. This is the main goal of phishing scams, although the term "phishing" has no clear definition due to the ever evolving nature of scams on social media platforms and the many forms each type of scam may take. Some definitions indicate that only replicated websites are a possible medium to conduct such attacks [7]. Marwe (2005) defines phishing as a "fraudulent activity involving the replication of an existing page to deceive users into willingly submitting sensitive information" [7–9]. Phishing scams are classified by multiple governing bodies as one of the lowest occurring scams equating to only around 5% overall but as the 2023 to 2024 period ranked among the most profitable scams on social media platforms [25, 73]. Current technology enables the perfect replication of websites with minimal effort through the use of multiple AI applications which brings the increasing need for an application to prevent this type of scam before the user accesses the malicious sites [73].

Romance Scams

Through the global communication and open information that social media platforms provide, romance scams have embedded themselves as one of the most prevalent types of scam among the older generation and individuals with limited offline social connections [22, 32]. Romance scams entail a cyber criminal using a fake persona to establish a romantic relationship or emotional connection with the victim through social engineering techniques for financial gain. Scams of this type aim to initially develop this connection at a fast pace in order for the cyber criminal to evaluate the initial effectiveness and determine the long term possibility of the financial benefits [22]. Eventually, cyber criminals

communicate an emergency that requires immediate action and request funds from the victim to help resolve the hoax circumstances. Once a victim has given into the requests they are often groomed for long periods of time through text communication only and scammers will always look to develop an excuse to never move onto any other form of communication. Urgent solutions are required for scams of this type as they rely heavily on communication through messaging applications in which user conversation data is not monitored to uphold privacy and governing policies.

Impersonation Scams / Identity Theft

Impersonation is used as the basis for many other types of scams as it provides a veil for the cyber criminal to use in order to hide their identity and capitalize on pre-existing connections or trust that the victim may have with the entity [49]. Impersonation scams take on many different forms with scammers posing as legitimate organizations, individuals, or friends and family either through cloning or hijacking social media profiles. It allows for cyber criminals to minimize suspicion and exploit social trust in efforts to extract funds or sensitive information from the victim and the most commonly impersonated profiles on social media are high profile people and commonly known organizations with it either being the organization itself or a person employed by them [24, 49]. Additionally social media platforms allow impersonation scams to thrive via the exploitation of minimal moderation around accounts and advertisements while also relying on the trust users have established with the social media platform itself or its content. Studies from the Australian Competition and Consumer Commission (ACCC), US Federal Trade Commission, and GASA conclude that impersonation is the highest contributing factor to the overall financial loss due to scams with impersonation scams now being considered the umbrella term other scam activity [5, 10, 11, 26].

Scams of such as those stated above and many more are the reason a proactive and real-time solution is required to identify online scams and evolving scam trends. Without a solution that is capable of evolving alongside the ever changing world of scam, statistics show that the growing financial loss will continue to grow. Additionally current technologies in regards to scam prevention are focused on educating users to improve personal judgment rather than developing a proactive method for collecting data on current scams and working towards implementing solutions.

2.2 Data Collection

Data collection is the foundational component in developing evolving AI-Agent scam detection systems for scam activity on social media and communication platforms. Data collected through multiple processes such as web scraping, crawling and Open Source Intelligence (OSINT) gathering server multiple purposes throughout the operations of AI-Agent based systems as well as the development for systems in the nature of scam prevention. Firstly, initial data collections provide the foundation for training and fine

tuning core Machine Learning (ML) components building scam detection focused AI-Agents. Secondly, following datasets and continuous data collection enables AI-Agents the ability to adapt and evolve alongside the ever changing world of scam. Through progressive model tuning, refinement and Retrieval-Augmented Generation (RAG) agents are no longer stagnant from the time of development within their key roles [29, 35, 50, 68, 69]. Finally, the data collected throughout system operations is the core foundation for the resulting outputs of cyber-security systems.

The process of data collection for cyber-security systems and scam detection on social-media/communication platforms is achieved through a combination of multiple techniques and technologies. To meet the "big data" requirements needed the process of web crawling, scraping, and OSINT gathering via any input all need to be utilized in tandem to gather data from all internet sources available. Anne L. Buczak et al (2015) highlights that big data is a critical component for key cyber-security system components including Data Mining (DL) and Machine Learning (ML) and RAG implementation [15, 44]. Anne L. Buczak et al. also states that the quality and quantity of data will effect model performance, system operations, as well as final system outputs. This is validated through literature such as that from Marinho er al (2023) which further state that the collection of data regardless of its correlation to the desired system output still provide support for AI and Generative AI functionality [15, 44].

Scam related data within social media platforms encompasses user profiles, groups, pages, posts, comments, and shared information among platform users, however this is not the only data required to counteract scam activity on the internet [33, 45]. Alongside the platform specific data collected, external data gathered from other sites plays a pivotal role in disrupting cyber criminal organizations as this is where data related to scams conducted over communication applications is shared. Additionally, the externally sourced data enriches that which is platform specific by providing further context and relation to other critical pieces of scam related data [51, 56]. Scammers frequently user social media platforms as an avenue to reach potential victims but also employ further methods of trapping victims by contact through communication platforms where they will then lead the victim to external sites such as banking, phishing, or false shopping sites to carry out their fraudulent schemes [4, 8, 62]. Consequently, data collection aimed at detecting evolving scam activity on social media must extend beyond platform specific boundaries, as additional contextual information is needed to ensure a high level of accuracy in the conclusions drawn by AI-agents during the decision making process.

While the process of data collection is important for AI-Agent agentic cyber-security systems to counteract scam activity on social media/communication platforms, it is also used by scammers to source potential victims and conduct online scams. The benefits data collection provides for scam detection also allow cyber-criminals to streamline the process of identifying potential victims, setting out honeypots to trap people, and gather information regarding currently identified scams [45]. For instance, victims are identified

if there is a large enough exposure to personal information that can be used for an impersonation scam over the phone to banking or government organizations [54]. This is just one example of how scams have evolved over time to incorporate data collection techniques into the criminal pipeline, replacing previous methods such as mail theft to collect sensitive victim information. Additionally, the digitalization of data collection techniques with AI integration has allowed scammers to expand their reach to a global scale at a rate faster than previously possible [4, 40]. This is a key factor in the rate at which scams continue to evolve and demonstrates the need for real-time monitoring of large scale data at a high volume to provide anti-scam systems with the same rate of evolution.

Cyber-security agentic AI-Agent systems in the field of scam prevention are commonly integrated with dedicated knowledge bases such as "MITRE ATTACK" and other OSINT platforms to ensure that enough data is collected in relevance to new and emerging scams. MITRE ATTACK is a large scale OSINT organization that shares an open source knowledge base of attack behaviors organized in a way that allows for the understanding, detecting, and tracking of evolving cyber threats or scams [33, 44]. This data is required to establish a foundation for scam behavior classification but scammers have employed methods to use the same openly available information to evolve their scam tactics at a rate faster than organizations like such can adapt. These curated tactics developed around the openly available scam behavior knowledge are extremely effective as users are given a false sense of security if the scammers tactics don't align with the out of date information given [4]. Scammers have also employed generative AI to curate fake intelligence to be fed into these systems, hindering the accuracy of the knowledge base or altering its analysis at time of release [68]. Considering these factors, it is evident that an agentic cyber security system developed for scam prevention must use its own internal knowledge base alongside those that are publicly available to effectively evaluate data and dictate its own path in detecting and tracking evolving scams.

Data Cleaning, Pre-Processing and Feature Extraction

Data processing is mandatory to successfully train the internal machine learning algorithms and deep learning modules in AI-Agents that give them their ability to core functionalities. Cleaning and pre-processing the collected scam data is a vital step in ensuring the raw data collected from various sources is clean, consistent, and suitable for analysis [50, 68]. Handling of missing data and outliers in training datasets is critical to providing agentic AI-Agents with the foundation to effectively evolve alongside scam activity and produce accurate real-time insights [69]. Within literature, Gumphusiri et al. (2024) proposed both a single agent and multi-agent method for data generation for use in training deep learning architectures with the purpose of conversational classification in scam detection. It was concluded within the research that training datasets developed via generative AI (specifically LLMs) achieved higher classification accuracy in comparison to real world datasets [29]. This higher classification accuracy can be attributed to the

lack of real world data with correlating features to ensure the outcome of models trained in scam detection to achieve an acceptable level of accuracy [29, 68].

2.2.1 Open Source Intelligence - OSINT

Open Source Intelligence (OSINT) gathering is the process of collecting publicly available data, either manually or through manual input, to curate a public knowledge base. OSINT alleviates the limitations of collecting data from a single source, expanding the overall scope and amount of data collected to be provided to AI-Agents within anti-scam systems, allowing the capability to draw correlations in the analysis process [44]. Pastor-Galindo et al (2020) states, the ability for artificial intelligence to make complex correlations that are naturally unpredictable to humans, as the key to producing accurate conclusions closer to the desired goal [56]. The enrichment that these correlations provide to the data is pivotal to the adaptability of cyber-security systems if they wish to adapt at a rate matching the evolving landscape of scam activity on the internet. An example of OSINT gathering in the case of social media scam detection is the tracing of phishing URLs shared in platform content to gain additional data needed to identify scams across multiple domains [4].

In contrast to the benefits OSINT provide, there are implicit drawbacks of additional complexity in data management, cleaning, storage, and processing. In efforts to minimize these impacts, the previously explored literature revolving around data processing is required alongside the training of ML/DL based AI-Agents dedicated to verifying source integrity [44, 60]. Social media and communication platforms also pose significant limitations to the effectiveness of techniques used for OSINT gathering due to the restrictions of publicly available content on these sites and the hiding of sensitive profile information behind user authentication. Platforms such as Facebook, WhatsApp, and TikTok impose these restrictions as apart of their efforts to reduce external entities from employing data collection techniques on their platforms and protect user data [27, 33]. Due to these limitations, single platform anti-scam systems are hindered through the inability to collect the critical data needed for the classification of scams and identification of scammer profiles and content [27].

Open source intelligence gathering is the primary form of data collection adopted into the scam activity pipeline of cyber criminal organizations. In contrast to how OSINT gathering is used in cyber security, scammers use the same techniques to collect user data from external platforms to assist in the victim identification process and tailoring specific scams to the target for maximum efficiency before contact is made [33]. Conducting data collection for means as stated also provides scammers with additional benefits to obscuring their malicious intents because it removes the need for them to solicit information from the target that was previously required in the scamming process. This demonstrates how OSINT is considered to be an effective tool in both scam activity and scam detection for social media platforms, even without the ability to bypass user authentication restrictions.

As this is the case little literature covers the intentional poisoning of OSINT by scammers in efforts to poison AI-Agent cyber-security scam detection systems that use mass OSINT databases for training of specific models [33, 40, 60]. While referenced as a possibility it is not explored as an anti detection technique used by scammers on social media or communication platforms. As previously discussed scammers are able to benefit from OSINT used at a basic level while its integration into more complex systems can cause negative effects on the overall results.

2.2.2 Web Crawling and Scraping

Web crawling and scraping are the technical means for automating the search and extraction of data from the internet. Web crawling is the method of traversing links or pages, while scraping is the extraction of data from the web pages accessed. Leveraging the combination of these techniques is a critical component of OSINT knowledge bases and select cyber-security platforms. With the conjunction of AI-agents the process of data collection is a critical component in scam detection systems for social media and communication platforms. There is little literature around the use of specific web crawling or scraping techniques and tools within the context of data extraction for cyber-security systems due to the importance of the data collected itself rather than the methodology used to collect the data. This is the case because there is a vast variety of different technologies used across the industry that can achieve the same results needed for these systems due to their many shared capabilities. An educated evaluation of specific strengths and weaknesses for specific tools will be used to highlight the capabilities and relevance to the use of AI-agents for scam detection and data extraction in the context of social media in order to demonstrate how specific frameworks are more suitable for social platforms. In conclusion to this fact, data can be collected in many different ways through web scraping, crawling, OSINT gathering, manual collection, etc, but it is this data that holds value to AI-Agent systems, and while some tools may be favored, their use does not determine the data quality.

Within the context of social media and communication platforms there are two main technical stacks that are predominantly used within the industry. The first technological stack is Python based, where the combination of the Playwright library is used for the web crawling process and the BeautifulSoup library is used for scraping and HTML/XML passing for data extraction. This specific python language based combination is used for its ability to handle a wide range of pages while also being able to process dynamically loaded and JavaScript heavy web pages [20, 48]. The second technological stack is JavaScript based and is the combination of either Playwright or Puppeteer with the Cherio or JSDOM libraries for the web scraping component [20]. In addition there are many other frameworks and libraries that are capable of these tasks, such as Selenium or Scrapy for web crawling, but fall short in functionality for certain goals when compared to those previously mentioned. To meet the requirements of a multi agent cyber security system designed for scam detection, a combination of both the Python and

JavaScript/TypeScript language and their respective frameworks/libraries are required to benefit from their strengths and alleviate the weaknesses of others such as the handling of dynamic data loading infinite feeds on social media platforms [48].

Web Crawling is where both Python and JavaScript provide a multitude of frameworks for data extraction and while Python is favouritised for doing so, JavaScript is often integrated to handle the dynamic loading of content which can then be passed back to Python frameworks for extraction. The Playwright framework stands out for its ability to achieve this, as is considered vastly superior in handling modern JavaScript heavy sites with its applicability for use with both languages. Developing AI-agents for the purpose of crawling and scraping data across the internet with a combination of frameworks as such provides these dedicated agents with the tools needed to ensure that the system in its entirety faces no limitations [13]. This approach ensures that the necessary data for detecting and tracking evolving scams on social media is collected with supporting data from external sites providing the enrichment needed to achieve the desired level of accuracy [20, 48].

There are a vast variety of different cyber investigation platforms and frameworks available to integrate into scam detection pipelines to assist in the process of collecting data alongside web scraping. Some of these frameworks such as Maltego or Spider-food already have AI integration capabilities but without a truly dedicated AI-agent that is capable of adapting to evolving trends within scam related data, systems that use these tools as standalone components will stay stagnant [43]. Social media platforms also require user authentication for critical data collection, a key flaw in some frameworks as they are unable to complete authentication bypass tasks due to limitations and specific ethical considerations. This is why modern web scraping tools and techniques must be used in conjunction to openly available frameworks for AI-agents to given dedicated perception tools that also provide the ability to be used for actuation.

Social Media Anti-Scraping Techniques

In efforts to counteract the collection of data from social media and communication platforms, organizations employ a variety of anti-scraping techniques in efforts to disrupt traversal of their platforms and extraction of data. Commonly, platforms are designed to generate web content dynamically as this makes the development process easier but also creates an environment that is extremely difficult to analyse. Dynamically loaded content such as infinite scrolling feeds and header generation is also commonly used in efforts to hide "ghost fields" that act as honeypots for the platform to catch automated systems and if triggered will restrict user access [13]. "Ghost Fields" are hidden JavaScript or HTML elements within a page construct that are not visibly displayed to the user. This is the case because to a web crawling and scraping application that transverse page elements through component analysis, these hidden artifacts are actionable items. If interacted with these honeypot style traps lead to rate limiting and more commonly IP

blocking, essentially halting system functionality and remove them from the platform. Anti-scraping techniques utilized by social media and communication platforms are considered a double edged sword in regards to the world of scam. For cyber security based systems they heavily impact the core functionalities that are designed to assist users but also limit the capabilities of scammers searching for sensitive user information in the same way. Although when considering these limitations scammers often have the advantage as they employ manual OSINT gathering techniques to identify victims with high levels of exposed data that is not hidden behind platform security measures.

2.3 Artificial Intelligence in Cyber Security and Scam Prevention

Artificial intelligence (AI) has rapidly evolved into an indispensable component in addressing the growing challenges around cyber security and evolving scam activity on the Internet. Within the realm of scam prevention on social media and communication platforms, AI enhancements through agent integration have been applied to nearly all core components, including content moderation, advertisements analysis, user authentication, and platform behavioral analysis [4]. Facebook, WhatsApp, TikTok, Instagram, and many more have heavily integrated AI driven solutions in efforts to counteract the proliferating fraudulent content distribution, such as investments scams, phishing sites, false advertisements, etc, through their platforms [3, 4, 30, 40, 51]. The impact of AI in this way is visibly profound with a stark difference between previous iterations of platforms to their current states. Although these changes have significantly changed the industry with moving towards a safer environment for users, the nature of scams is evolving at an incomparable rate and causing many AI integrated security/moderation features to be ineffective by the time of release [54, 74].

Advancements in Machine Learning (ML), Deep Learning (DL), and Large Language Models (LLM) have enabled the development of complex multi-agent platform specific security systems capable of data mining extensive amounts of data to extract deception patterns and anti-social engineering trends from malicious activity [28, 55, 56]. General understanding through high level overviews present that Natural Language Processing (NLP) through Neural Networks, Large Language Encoder-only models, and encoder transformers are used for classification of user content which is then compared against tightly controlled ML system outputs to identify outliers [61, 67]. These outliers are what is considered to be unwanted interactions where measures are then taken to remove them from the platforms. Little literature is published regarding these systems as the information is considered extremely valuable to organizations and could also cause leading negative effects if those with malicious intent were to understand their processes and develop methods to circumvent operations [37].

Despite the little literature regarding the advancements of AI in social platform cy-

ber security systems, there is literature to support significant limitations regarding the accuracy and bias of moderation systems when trained on limited or pre-empt bias data specific to a single platform [34, 50, 58]. Laurence Moroney et al (2020) highlights the importance in extensive training/fine-tuning of ML, DL, and LLMs with the use of large data quantities specifically crafted with diverse data correlations to achieve un-bias high level model accuracy [50]. This is further supported by academics Paul Lusztin and Maxime Labonne (2024) who emphasis utilizing clean and valid data that is diverse in nature for the model training processes of advanced AI-agents as they are inherently susceptible to noise if not done so [34, 58]. There exists a significant disparity between these stated claims and the practical implementation of AI-Agents in cyber-security systems designed for scam detection. The nature of social media and communication platforms to allow users open communication in conjunction with the continuous spread of misinformation will never produce an appropriate single platform dataset that can be used to train an AI system to a satisfactory level of accuracy [38, 42, 54].

Although the environment for training systems to meet an acceptable level of accuracy may be unfavorable, when considering AI-agents in the nature of cyber-security for the prevention of malicious activity, a level of acceptable bias or domain-informed bias (in favor of pretesting the user/preventing harm) can be considered beneficial. Human augmented training datasets are commonly used in fine tuning models for the purpose of adding preventative measures against false negatives [38, 53]. This level of acceptable bias is demonstrated in similar cyber security systems such as deep-learning based anti-virus models where resulting outputs are provided as suggestions to users instead of actions. This is contradictory to the actuation component of AI-Agents but in the case of protecting users, limiting the resulting effects of false negatives and mistaken data interpretations is considered favorable. When drawing a comparison between the growing yearly financial loss due to scam activity and the awareness of AI threat capabilities indicated in Figure 2.3, one can draw the conclusion that a level of bias in favor of caution can be considered favorable if leading towards lowering the impact of scams [11, 25, 26, 53, 64].

2.3.1 Generative AI

Generative AI is the classification of AI models that learn data correlations and synthesize new content, either being test, images, audio, video, or code. Commonly in the world of scam there are three key model uses, encoders/decoders for text understanding and generation, diffusion models for video/image creation, and auto-regressive models for contextual understanding. Within the little literature available regarding this topic, the specific use of large language models for the role of scam detection in content and scam content generation is explored to evaluate the overall effectiveness [19]. Chan et al. (2024) and other academics conclude that from their findings, the use of such generative models are ineffective in producing acceptable outcomes that are true replications of human interactions, yet fail to mention that these test cases we conducted within single platform environments using closed datasets for testing and model training. When compared to

literature that covers the use of multi-agent deep reinforcement learning (MADRL) for cyber security, results dictate that an approach taken using specific architectures is extremely effective for both classification of content and content generation over long context periods [70].

Large Language Models

Large Language Models are the specific sub-category of generative AI that is most commonly known to be employed by scam entities online to generate content and perform tasks that streamline the scamming process. In the context of scam content generation specifically, the user of single agent systems for the purpose of content generation is extremely effective as LLMs are able to maintain a context window that is suitable for generating what is required by scammers to employ their newly adopted tactics [37]. LLMs are most commonly deployed into the scamming pipeline as they are easily accessible and provide high user simplicity making them effective for quick actioning with user input to craft text specific to the situation. Current statistical reports conducted online by global anti cyber-crime organizations demonstrate that there is a growing awareness around the use of generative AI and specifically large language models in current scams that potential victims have had contact with.

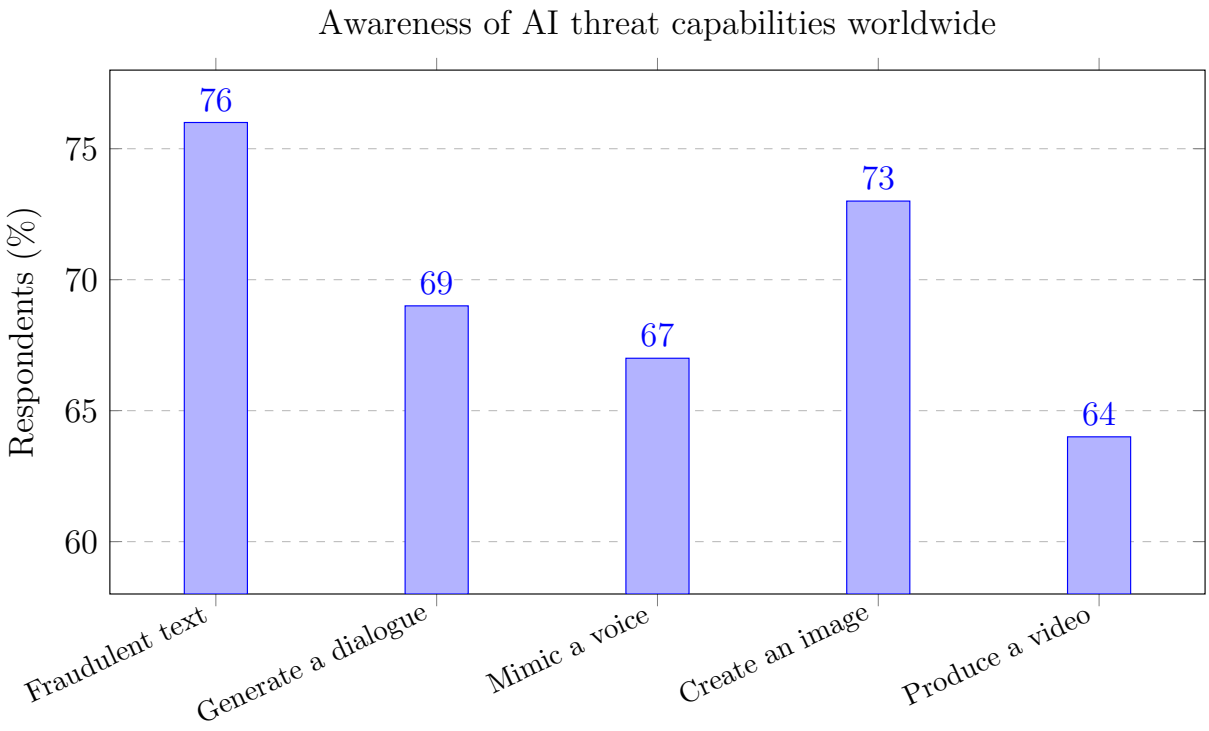


Figure 2.3: Overall awareness of AI threat capabilities is increasing worldwide - GASA [25]

In Figure 2.3 it is presented that a high number of respondents where able to indi-

cate that content they had interacted with was generated using artificial intelligence, yet within the same reports it is also presented that there is a growing financial loss due to scam activity online [25]. Little literature explore or support the growing issues regarding misuse of AI within the context of scam media and there are none offering a solution to counteract the efforts of sharing this content either through social media or communication platforms. Specifically, communication platforms such as WhatsApp, Telegram, and Messenger are the main channels where AI generated text is used and where LLMs are employed to maintain conversations with victims as they are able to generate text relevant to the contextual window they hold [37, 57]. Detecting the use of such tactics to employ scams such as online phishing through these communication platforms is an area where exploration is growing due to the high demand for a solution that is capable of preventing malicious activity before action from financial organizations is required [37, 57]. Large language models are increasingly being used on communication platforms to spread scam material for their Natural Language Processing (NLP) capabilities and their refined ability to understand indictions that traditional systems overlook, providing context aware sophisticated evasion techniques [37].

2.3.2 AI Agents

Artificial Intelligence Agents are systems that perceive, plan, and act within an environment towards a goal while adaption overtime [66]. The cyber security systems in place by current social media platforms for moderation are ML driven agents designed to uphold a specific rule as their main goal. Effective in achieving a specific task such as language enforcement or other general platform interaction guidelines, these type of AI agents are unable to adapt within the complex environment of scams on social media, constantly being left behind as scammers change their tactics to avoid punishment. For this fact, modern LLM-driven agents are required to provide a systematic solution that is capable of adapting at a rate consistent to its environment, specifically for their superior use capabilities, planning, and self-reflection [39]. Evolving cyber security systems in place for specific goals such as filtering common phishing email scams, utilize the LLM driven agent architecture for the purpose of being able to implement a multitude of self-reflection and learning systems from their own input data while in operation [39, 66].

Social media and communication platform scam security systems have little literature exploring their functional operations or design due to the new and developing nature of the environment and research. Comparative to cyber security systems such as anti-virus systems, this is due to not wanting to provide malicious entities with the inner workings of these systems so that malicious entities know how to produce content that is capable of avoiding them. From the literature that is available, a common conclusion is that the incorporation of LLM based models within AI agents for being able to understand context and human behavior is required [4, 55, 66]. Agents capable of NLP with contextual understanding are able to plan accordingly to changes in the systems deployed environment and appropriately response with the according provided tool [39, 55, 66].

Retrieval-Augmented Generation

Moderation based cyber security mechanisms incorporated by social media and communication platforms utilize the process of Retrieval-Augmented Generation (RAG) in provided system prompts to add human controlled restriction. This is achieved as RAG implementations using human developed example datasets with specific data labeling and relations provide controlled goal clarity [40, 55, 66]. RAG specifically is where a generative model is provided a retriever that provides relevant data from an external knowledge base, and it is this factor that modern agent implementations use to add human controlled guidance/direction. Scams on social media are an ever changing environment and with privacy restrictions around personal user data on communication platforms, assistance is required to contextualize the environment for where an AI agent will operate. Most notably used by some OSINT agencies where AI-Agents have been integrated into the web scraping process to improve efficiency and use tools to provide context to the page data with NLP [14].

2.4 System Architectures

System architecture in cyber security systems follow a common practice depending on the desired goal, in the category of AI Agent centric systems there are two main design types. The first of these being an open loop feed forward architecture adopting a streamline or pipeline process used by social media organizations for moderation systems as they follow a simplistic data direction and similar design to goal philosophy [66]. Data is collected and feed into one end of the pipeline and then through a series of data labeling, analysis, planning, and then finally action, a result is produced, this is often in the form of account suspension or warning. The second is a closed loop or orchestrator architecture that employ a system feedback loop to support system progression, commonly seen in anti-virus software [1, 66].

LLM centric AI Agents perform optimally when deployed in orchestrator architectures due to being able to specialize in a dedicated task and be provided a controlled environment to perceive changes. Limitations placed on AI agents deployed in feed forward architectures are considered minimal in orchestrator architectures as LLMs are fined tuned to impose push-back and re-evaluation on actionable decisions before a final outcome is concluded. While this may be the case, the use of multi-agent architectures does not circumvent that artificial intelligence applied in single environment social media or communication platforms will demonstrate bias and inaccuracy tailored to the platforms content. To elevate this limitation a combination of architecture design and multi-platform big data is required to truly grasp the variation in content and context that must be analysed to produce reasonable outcomes [38, 42]. This is further validated as modern moderation systems on social media platforms and specifically communication platforms continuously produce lower accuracy results in standardized quantitative tests, while multi-agent systems following a cyclical data flow orchestrator architecture

demonstrate that self reinforced learning and continuous evaluation outperform others over longer periods of time [1, 55, 66].

AI Agent architecture foundations are constructed on the core principles of perception, where inputs are perceived as a change in the environment they operate in, reason, where the planning of action is conducted, and finally actuation, where a final action is taken in response to changes in the environment. The self-reflection component in AI agents is tied into the perception and reasoning stages of the system thought process, ingested inputs are contextualized through RAG or goal re-enforcement in the perception stage and then feedback is provided to support reasoning and execution planning. This core principle is what defines an AI agent, yet the means of achieving these internal processes may differ based on design, development and implementation. In the specific instance of specialize AI agents, shared knowledge bases are used during the perception process to add context to data and then again in the reasoning process to support action planning.

2.4.1 Scam Prevention Technology

Literature exploring scam prevention cyber security technologies available in the industry today is scares due to the fields early development stages and industry focus on secrecy. The most notable cyber security platforms available today share no information regarding their processes, methods, or implementation but provide a general outline of functionality and operation. Additionally these systems are designed with single focus tasks in mind, such as the 'MasterCard Consumer Fraud Risk' platform that monitors consumer usage for suspicious activity that differs from consumer behavior. Similar non-enterprise based cyber security systems for scam prevention such as 'FeedzAI' or 'SEON' focus on transaction or system monitoring and only provide alerts to threats once they have been noticed rather than providing users with preventative measures. Although this is the case they each share many similarities, through monitoring user behavior and then comparing the outcome against user behavior so that any differentiations are alerted to the user. It is this focus on measures taken after that fact of third part actions that is why a real-time solution to scam activity on social media and communication platforms is needed to act before malicious entities are able to conduct their operations.

Chapter 3

System Overview

The following chapter provides an overview of the proposed system designed to produce an adaptive real-time solution to scam activity on social media and communication platforms. The solution will operate as a stand alone system with monitoring integrations into social media environments and communication platforms. The primary goal of the system will be achieved through the successful operation of a multitude of AI-Agent collaborations. A search for scam material is made on the connected monitoring platforms as well as the wider internet, scam related data is extracted, this data is then verified and classified into scam categories. An analysis is conducted on the resulting data to produce actionable trend intelligence and conclude data correlations which will be utilized by the overriding orchestrator to control system operations. A breakdown of the system requirements that must be meet is provided, alongside a description of the two system architectures used to design the system. The systems developed each respectively follow an architecture and a comparison will be made between their performance in order to evaluate system efficiency and effectiveness in progressive self learning. In this chapter a high level module description is also provided to explain the core principles of each key module that will make up the inner workings of both systems, and each architecture will incorporate specific core principles through different implementations.

3.1 System Requirements

In determining the criteria for system requirements, a comprehensive analysis of social media and communication platform scam activity was conducted. Alongside this information, scam related statistical information from governing bodies, LLM capability evaluations, and comparative AI Agent design methodologies was used to construct the core agent principle requirements. The proposed solution is designed to meet the following functional, non-functional, and interface requirements to ensure system operation goals are meet. The requirements are also designed with consideration to align with the continuously changing landscape of social media and system goal in providing an adaptive solution to the problem of internet scams.

Functional Requirements

Fundamental requirements provides the necessary features and functionalities of the system to achieve the project objective.

Requirement ID	Requirement	Importance
F-ReqID-1	The system will collect data from both public and social media domains in real-time	High
F-ReqID-2	The system will bypass anti-scraping measures imposed on social media platforms	High
F-ReqID-3	The system will process and filter collected data in real-time	High
F-ReqID-4	The system will analyze processed data to identify and classify potential scams via the employment of LLM-powered AI-Agents	High
F-ReqID-5	The system will produce scam trend analysis reports at scheduled intervals via effective prompting of LLMs	High
F-ReqID-6	The system will adapt to patterns identified through data analysis	High
F-ReqID-7	The system will store data collected and the analysis reports generated	High

Table 3.1: Functional Requirements.

Non-Functional Requirements

Non-functional requirements specify the goals required to achieve quality and optimal performance.

Requirement ID	Requirement	Importance
NF-ReqID-1	The system will maintain maintain a continuous up-time	High
NF-ReqID-2	The system will report an overall trend analysis of its operational lifetime when prompted	Medium
NF-ReqID-3	The system will scalability to handle cross platform analysis and increased volumns of data	High
NF-ReqID-4	The system will record all prompt responses for maintenance	High

Table 3.2: Functional Requirements.

Interface Requirements

The interface requirements outline the goals necessary for effective communication between the modules within the multi-agent system

Requirement ID	Requirement	Importance
I-ReqID-1	The system will provide API endpoints for external integration of other applications	Low
I-ReqID-2	Agents must communicate via asynchronous message passing following the FIPA communication structures	High
I-ReqID-3	Agent communication channels will securely transfer data between components	High

Table 3.3: Functional Requirements.

3.2 System Architecture

This section will outline the two system architecture types being developed as apart of this thesis, the first follows an open loop feed forward pipeline and will serve as a baseline for when testing progressive system and agent learning through internal system feedback loops. The second architecture follows a closed loop feedback agent architecture with an additional orchestrator agent to provide the system with its internal self learning processes. The following section will outline the key differences in system operations for both architectures and how module implementation is to be conducted following the architecture type. This outline is to provide context for future test results and comparison of resulting metrics per architecture type in the following Results and Discussion chapters.

Closed Loop - Feedback Agent Pipeline Architecture

The closed loop feedback agent pipeline architecture is designed using a modular hub and spoke design type with a cyclical flow of data through the system and back to specific components. An orchestrator agent is implemented to provide feedback through specialized RAG support knowledge and goal adjustments through specifically crafted control variable implementations. This specialized knowledge will be provided through direct calls to the applications API rather than through specific queues on the message management system. Although the system layout is of hub and spoke design, data will be processed following the cyclical data flow outline and agent management communication will operate through the RESTful API implementations.

The Data collection agents will support external control adjustments to their search criteria and content targeting through endpoints that the orchestrator can interact with.

Both the scam classifier and analysis agents in the system will support specialized knowledge base example curation depending on the orchestrators understanding of future scam trajectories. As each agent itself will operate with internalized self reflection, the entire system following the closed loop architecture will replicate these operations to adapt and evolve alongside the environment it is deployed in. The closed loop architecture will be the focus for system design and implementation descriptions in the Methodology chapter and the following open-loop section will detail the limitations placed to produce the baseline system comparison results.

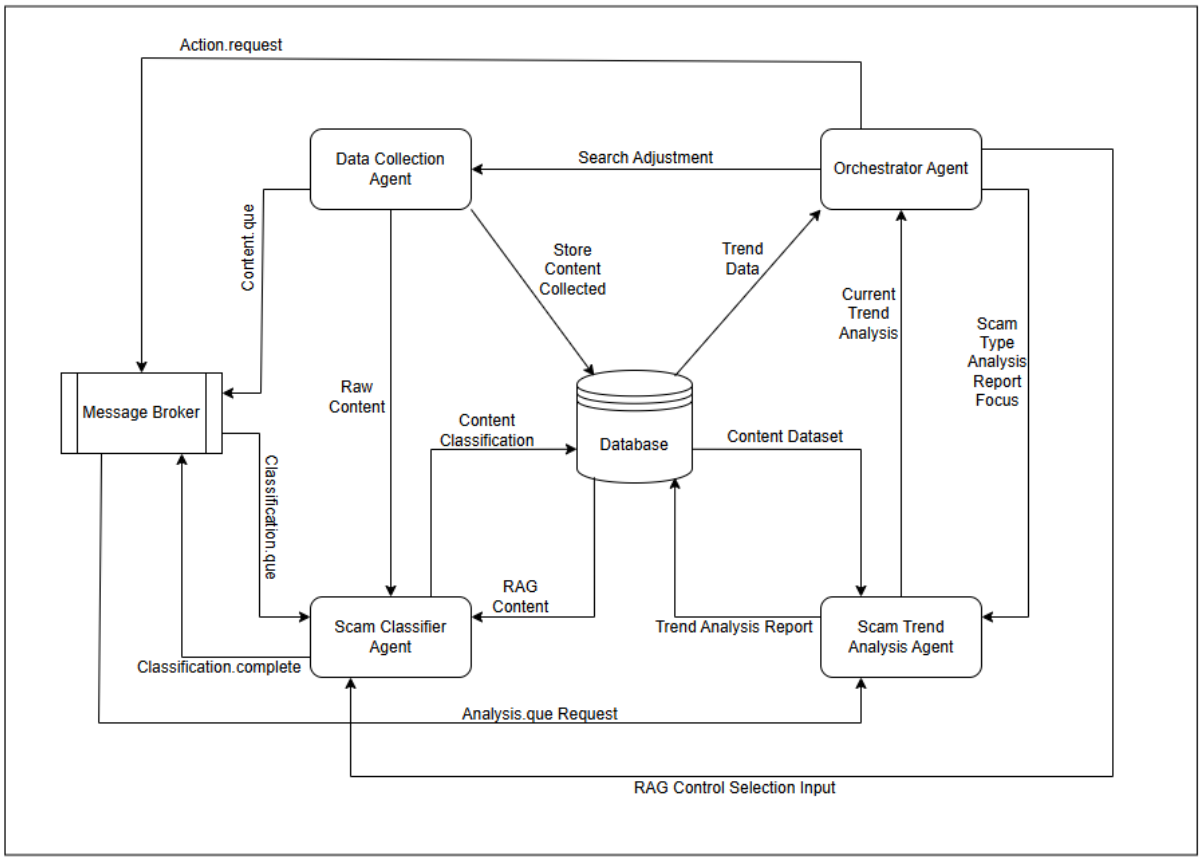


Figure 3.1: Closed-Loop High-Level System Architecture Overview

Figure 3.1 provides a high level overview of the closed loop system architecture with clear module outlines excluding the detailed integrations to external processes. The figure follows the proposed closed loop system architecture design implemented over a microservice based hub and spoke architecture type to provide clarity around module connectivity and system implementations using the selected system deployment services. Additionally displayed are the internalized system self learning loops that provide the proposed solution with the capability to adapt and learn from its deployment environment in real time. The internal self learning loops integrated into system operations are dedicated to individual agents, with the scam classifier agent and scam trend analysis agent working with content

retrieved from the database under the direction of the orchestrator agent. The outline of these self learning loops in Figure 3.1 are the data flow directionals from each agent to the database and from the orchestrator agent to the systems current agent in operation. The implementation of the system following this manner provides the complete solution to detecting and classifying scam content on social media and communication platforms.

Open Loop - Feed Forward Agent Pipeline Architecture

The open loop feed forward architecture is the most common type of system architecture when deployed for system moderation or any rule enforcement tasks. The utilization of this architecture in the project is to provide the purposed system architecture with a baseline comparison for evaluation and measuring the success of internalized system feedback/self-learning loops. Figure 3.2 provides a high level overview of the open loop architecture and data flow without the inclusion of the system orchestrator. The data collection agent can run multiple instances connected to the scam verifier and classifier agent, for testing purpose further explained in the testing methodology section only a single data collector agent will be run.

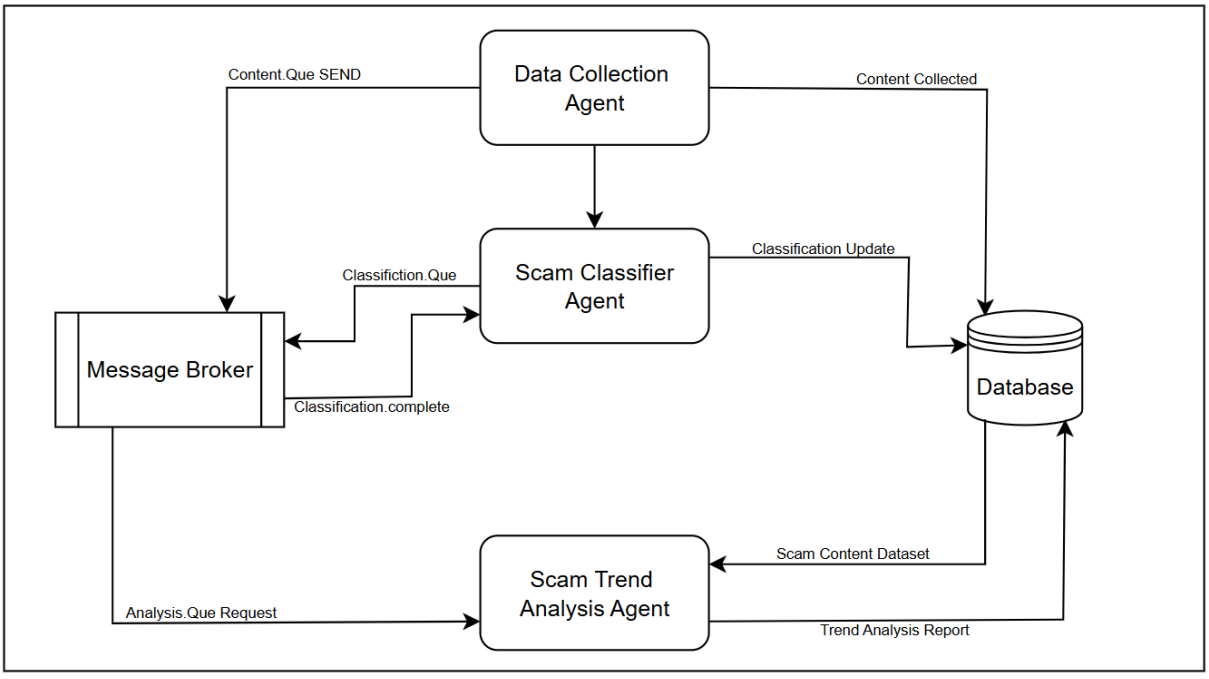


Figure 3.2: Open-Loop High-Level System Architecture Overview

The exclusion of the orchestrator agent is the only differing factor between the two architecture implementations and each agent was designed with an emphasis on modular design principles in order to be run in each architectural layout with no changes needed. To collect baseline testing results, Figure 3.2 presents the architectural implementation layout required with the exclusion of the orchestrator agent. All modules following this

architecture type follow the same design methodology and will be implemented following the same procedures outlines.

3.2.1 Module Description

Data Collection Agent - Web Crawler and Scraper

The data collection agent will operate to search social media and the wider internet to collect scam content and other social media material. This will be achieved using web crawling and scraping implemented using a combination of libraries and external APIs. Multiple instances of the data collection module can be employed to collect data from specific platforms or look for specific content it required to collect enough data to meet system requirements. Initial implementations of the data collection agent will operate using standardized controls set through environment variables to produce open loop architecture results. Further results from after the orchestrators implementation will be those following the closed loop architecture system design.

Scam Verifier and Classifier

The Verifier and Classifier Agent has two core functions, the first is to get the content collected verify the source to removing false positives collected from reputable sources, second is to classify the scam content collected into a specified scam category. The verification process will not remove data from knowledge base but mark it to be used for agent RAG processes. The classification process progressively uses categorized material to support further classifications alongside its provided rule based prompting. This AI-Agent will serve as the core module for system scam content progressive learning and the orchestrator agent will support this with classification adjustments. All key functionalities of this system will operate the same for both the open loop and closed loop architectures, with the only difference being the orchestrators ability to adjust knowledge selection.

Scam Trend Analysis

The scam trend analysis module is responsible for producing trend reports and conclude correlations in the data to group scams and display changes that are not noticeable by human review. This will be achieved using the Hierarchical Bayesian Structural Time Series (HBSTS) probabilistic model to learn trends, produce growth rates and predict short term trajectories. Resulting trends will additionally be stored in the knowledge base to be analysed using the same probabilistic model to assist the supervisor agent with orchestrating short term goals. The results from the scam trend analysis agent will be used to evaluate the effectiveness of adopting a closed loop architecture for scam detection on social media platforms. These results will naturally show a progressive difference in data over time if self-learning feedback loops have any impact on system outputs. Then following changes in results can be evaluated to identify the overall success and effectiveness when compared to a stagnant system.

Supervisor/Orchestrator

The Supervisor agent is the orchestrator within the system, providing the other modules with supporting functions such as RAG support to selectively provide information depending on requirements, and adjusting the other agents operations through variable and limit controls. The operation adjustments will be made based on evaluating the scam trend analysis reports and classifications from the knowledge base. The systems feedback and self reflection loop will be implemented through the interaction between the orchestrator agent and the other agents in this system. The open-loop system will operate without the orchestrator agent implementation and each module is designed to operate without implementation of the orchestrator agent to support this.

Chapter 4

Methodology

4.1 Introduction

The methodology chapter outlines the engineering centric approach employed to develop and evaluate an innovative AI driven multi-agent system responsible for detecting and tracking evolving scams on social media and communication platforms. This chapter systematically details the key development constructs as follows; the system architecture in its entirety and development purposes; AI-Agent design including a comprehensive breakdown of their functionality, methodology, and implementation; system technologies breakdown; AI model selection, use, and prompt engineering strategies. Finally in the chapter is the detailed system data communication documentation with storage, pre-processing and validation strategies. A design science research approach was adopted in the creation and assessment of the propose solution to address the research problems and goal to effectively detect and track evolving scams on social media and communication platforms. Quantitative objective metrics including accuracy, precision, and recall will be used as measurable outcomes to evaluate system performance and draw conclusions from system architecture comparisons for the final analysis and conclusion.

4.2 Detailed System Architecture

The proposed project solution is a Multi-AI-Agent system where several dedicated task components collaborate together to crawl the internet, scrape pages for scam related data, classify scam content, analyse trends in the data, and store information, with an overruling AI-Agent to direct communication, tasks, and refine system objectives. For architecture selection during the design process, the decision to follow a closed loop architecture for the multi-agent anti-scam solution was selected as it facilitated the desired functionality for system self-learning integration following a cyclical data flow. Agents in the system are designed to operate with and without the implementation of the orchestrator agent and its components to support both testing purposes and system modularity.

The anti-scam system was implemented using Amazon Web Services (AWS) cloud platform to host the Amazon Kubernetes Service (AEKS) cluster in which each agent is implemented as a single deployment. AWS was selected in the design process to host the system for the provided performance benefits in running a multi-agent real time system and which also directed the selection of Amazons Kubernetes service for operation simplicity. Following this additional design considerations where made for the selection of a Kubernetes cluster as future developments will require assistance from automated container orchestration, load balancing, and module scalability. Each agent on the cluster is a containerized module using Docker and hosted as an in depended pod with clearly defined resource limits and inter-service communication protocols. The system implementation places emphasis on modular design principles for scalability as the process aims to facilitate the ability to scale across multiple social media platforms during operation.

The system operates using its own dedicated database serving as the central knowledge base and data storage component in the system containing raw, processed, and classified data alongside analytical metrics. The database is implemented using PostgreSQL and hosted on Amazons Relational Database Service (AmazonRDS) as each system module incorporates the provided fail-over and retry support. Each module implements its own pooling and connection management through the Prisma ORM library where additional schema management is implemented on a module level. To support schema integrity the Prisma client also implements strict migration protocols and data typing.

Supporting the system is the message broker, implemented using a RabbitMQ deployment on the Kubernetes cluster which directs messages from one agent to another. RabbitMQ is utilized to support the real-time data processing functionality of the system with every item going through a specific order of processes before it is suitable intelligence. This data flow pipeline is dedicated to each new record found through by the data collection agent in the web crawling and scraping process. The dedicated message broken implementation was conducted using RabbitMQ to provide the microservice style hub and spoke system architecture with a cyclical data flow more closely aligned with the proposed closed loop architecture.

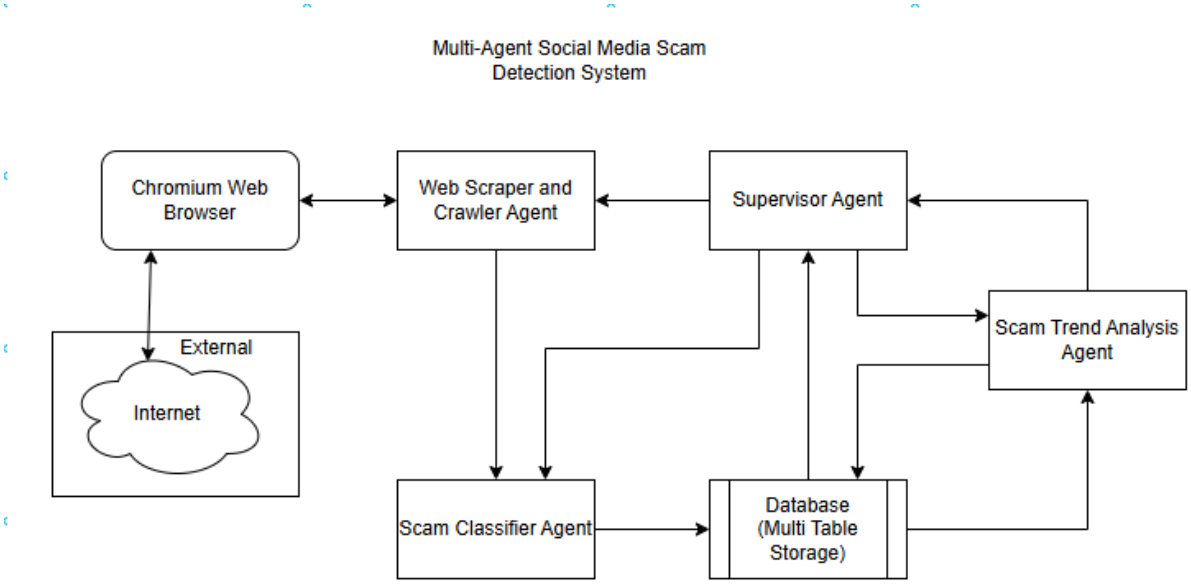


Figure 4.1: High-Level System Overview

Figure 4.1 depicts an indepth view of the system architecture in its entirety, core agent layout and system architecture components are displayed in black, with red used for the direction of data through the system in the real-time processing pipeline. Outside of the real-time data pipeline each module implements a standardized check for required record processing from the database directly. Additionally provided in Figure 4.1 is the core system API interactions that adhere to the architectural principles of Representational State Transfer (REST). Specific to module component implementations the data collection agent is a dedicated Python application wrapped in a Typescript RESTful API service and each of the other module implementations are standalone TypeScript Agents with integrated API functionality for direct module communication. The separation of both the RabbitMQ data pipeline and inter module communication protocols was selected to alleviate message queuing bottlenecks and data retrieval retry handling. Through the carefully crafted closed loop multi-agent system architecture built on top of a microservice hub and spoke topology, the complete system architecture is able to provide an adaptive system solution to detecting and classifying scam content on social media and communication platforms in real time when deployed in a continuously changing environment.

4.2.1 Agent Data Communication

How AI-agents within the multi-agent system communicate is a critical component to coordinating tasks while still allowing the individual agents to function autonomously. RabbitMQ a popular open-source message broker for asynchronous message communication is used as the middleware in the multi AI-agent system to decouple the agents in

relation to time. The defined channels needed for communication are based on the flow of data throughout the system and each agent is both a consumer and developer, with only the supervisor agent having a specially implemented channel for controlled agent self learning and evaluation. The core message channels of the system are:

Developer	**Consumer**	**Queue Name**
Data Collector Agent	Scam Classifier Agent	classifier.task
Scam Classifier Agent	Orchestrator Agent	orchestrator.dbevent
Scam Classifier Agent	Scam Trend Analysis Agent	trend.dbevent
Scam Trend Analysis Agent	Orchestrator Agent	orchestrator.report
Scam Trend Analysis Agent	Orchestrator Agent	orchestrator.dbevent
Orchestrator Agent	Scam Trend Analysis Agent	trend.generate
Orchestrator Agent	Scam Classifier Agent	classifier.task
Orchestrator Agent	Orchestrator Agent	orchestrator.eval
Orchestrator Agent	Data Collector Agent	collector.task
All Agents	Orchestrator Agent	orchestrator.record

Table 4.1: RabbitMQ Channel (Queue) consumers and developers in the Multi-Agent Scam Detection System

All published messages through the RabbitMQ message broker implemented in the Kubernetes cluster are stored in the database with an internal log of agent subscribers to each message channel encoded to the orchestrator agent. The implemented queue records table stores all transactional calls for the process of single content pieces and for dataset trend analysis reports. The record interactions table controls the internal system rate limiting of operations through a time to read and write value, artificially re-coupling the agents to an adjustable time limit imposed through the orchestrator agent. The record table is critical for control flow management while still allowing for the functional use of RabbitMQ's queue action and subscriber implementations on a per agent bases. Table 4.1 provides a complete outline of all message broker calls with each developer and consumer tied to the specific queue names. Each outlined call is dedicated to an actionable agent event with the inclusion of the success return message followed.

RabbitMQ has been selected in conjunction with a RESTfulAPI approach to allow agents to operate regardless of if a return message was received. The selection of this framework is suitable in the case of social media data extraction as it provides the ability for large amounts of data to be collected while the classifier and other agents work at the pace set by their LLM dependencies. While this is the case, a FastAPI will be established to the orchestrator agent for querying of system data, quantitative measurables, and human control to adjust either agent or system outputs based on developer requirements.

A design science methodology was used in the design process for the message broker queue and subscriber development as the broker ledger required careful curation specific to the proposed architecture type and data flow pipeline. The following module description and implementation section will provide additional detail to specific design methodologies and agent implementations for each agent in the system.

4.2.2 Data Storage

The proposed anti-scam system database serving as the central knowledge base is implemented using PostgreSQL hosted on AmazonRDS and managed on a agent level using the Prisma Client. A comprehensive design schema was developed to support the modular agent architectural design focusing on referential clarity and data organization.

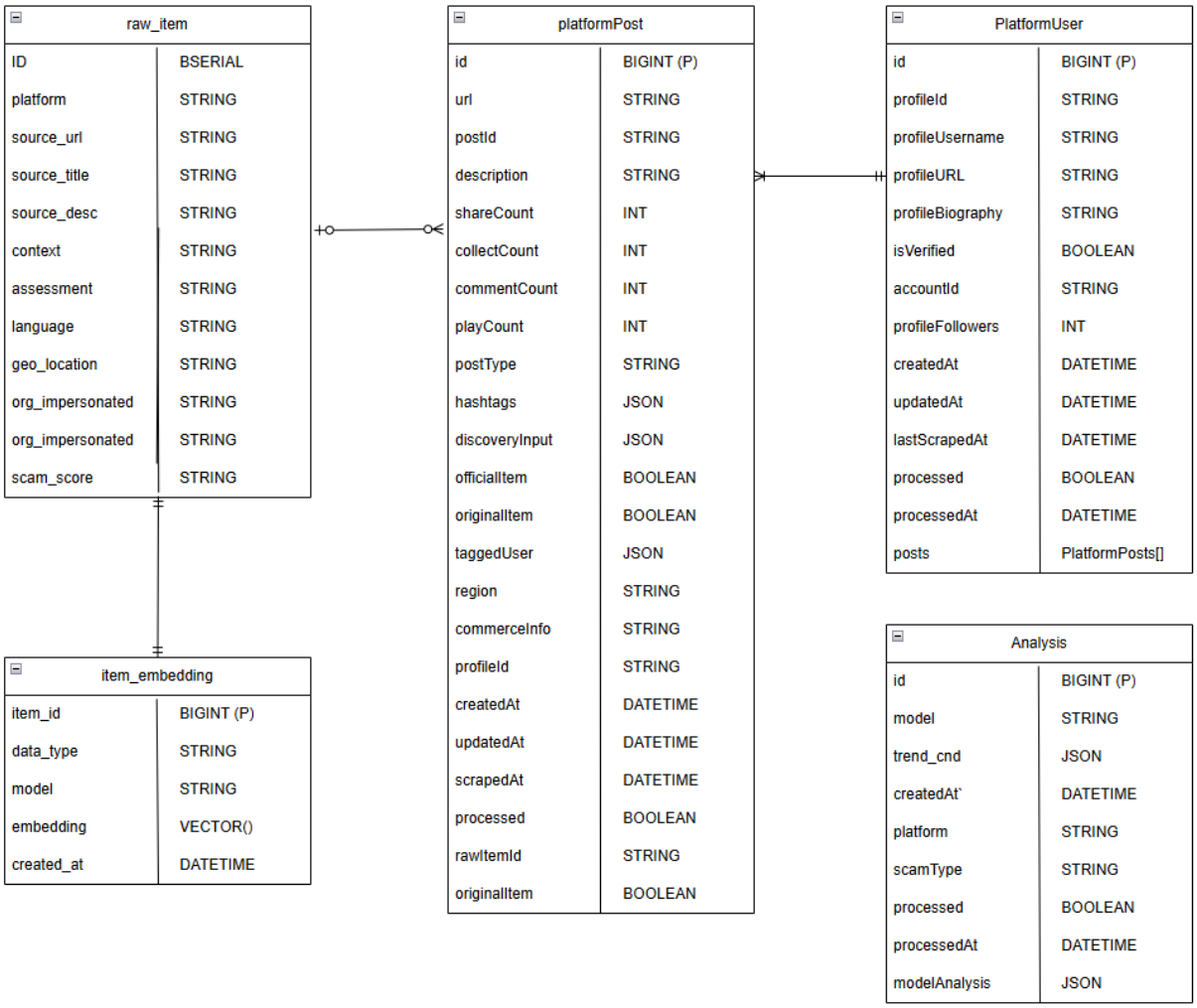


Figure 4.2: Database Schema

Figure 4.2 illustrates the database schema implemented in the anti-scam system, fol-

lowing four primary pieces of intelligence through distinct tables that form the foundation for data collection and classification. The fifth Analysis table contains the resulting output from the Analysis agent after processing a batch dataset collected from the four primary tables. The analysis table is responsible for storing the analysis report provided by the AI model, the trendcnd records stores the binary report string and finalized report data regarding the overall analysis at time of creating in a JSON file which can be read by the orchestrator agent or formatted and printed for human review. The table is a stand along table with no relational constraints to any other table, providing records as snapshots of the other tables without requiring data representation and for overall efficiency.

The platformPost and platformUser tables are implemented as the key tables for the data collection and classification agents in the anti-scam system, with a one to many relationship following an single user can have multiple posts. The AI tool call for the classification method uses both tables data as the knowledge base for retrieval augmented generation and prompt example support. The table implements both a primary and foreign key relationship to provide contextual information to each records and a general overview of the social content item. The data stored in both tables is entirely collection based features serving as the foundation for intelligence classification and solidifying AI generated features to real world content with context.

The rawItem table is implemented as the primary entity for AI generated features from the classification agent. The table implements an incremental primary key for each raw content item in potential scam content and a foreign key in a many to many relationship to relate AI capable data to a post for analysis. Following a modular design method the multi-relational partnership created through the rawItem table the database is able to facilitate linking AI generated features to a potential scam type. The assessment, org_impersonated and scam_score model generated features are appended to the raw_item data serving as human readable data records while the relational data stored in the item_embedding table is what the AI model processes. The item_embedding table stores the embeddings vector and model used for classification of a post through the AI analysis of its content and understanding of the numerically represented content.

To maintain data integrity all tables include a relational createdAt timestamp field for tracking generation time enabling the ability to track changes in content relations over specified periods of time. To ensure the integrity of continuously changing data records, the updatedAt and scrapedAt fields allow for progressive tracking in the data collection agent and ensuring that any changes made through one part of the system are consistently reflected in the connected data.

4.3 Detailed Module Design and Implementation

Following the system architecture outlined in section 4.2 this section outlines each AI-agent in terms of their role, design methodology, implementation, internal model use, provided tool calls, and API connections. Each agent operates autonomously while also under the overruling authority of the orchestrator agent designed to ensure the dedicated model methods operate in provided constraints and the following implementations demonstrate how the theoretical module designs translate to physical operations. Each subsection provides detailed implementation descriptions with respects to each agents core function, data processing workflow, feature generation, and external service integration.

4.3.1 Data Collection Agent

The data collection agent, implemented as an asynchronous web crawler and scraper, is responsible for two tasks, the traversal of external sources on the internet, and the collection of social related content data to be analysed for potential scam relation and classification. The agent combines both web scraping and crawling techniques with a pragmatic focus on potential data that is relevant to scam related activities. The combination of tools available to the Python language, being Selenium and Playwright, are used to transverse the internet and collect scam related data be effectively interacting and rendering dynamically loaded page content. The combination of these two libraries have overlapping features but allow for user emulated page control and infinite feed data extraction without the need for re-rendering or short-term data storage. Selenium in particular enables the AI-agent to handle user authentication and heavy JavaScript execution while simulating user interaction into the chromium headless web browser. The BeautifulSoup library is used for parsing the provided HTML and extracting the textual content alongside page metadata for the scam classifier agent. The combination of selenium and playwright provide the dynamic navigational ability of complex web pages that include infinite dynamic JavaScript loading without the limitations such as ghost fields interaction.

The web scraping agent will operate continuously in real-time and follow direction provided by the orchestrator agent, it will also be provided data extraction evaluations to assist in effectively self learning how to collect the necessary data for the classifier agent. The agent will operate with rate limit ceilings and multi randomized delays to bypass static social media moderation and security features. The multi-crawler technique will also allow for the agent to operate multiple accounts in avoidance of bot detection and anti-scraping mechanisms. Data extracted from the multitude of page sources will be structured and recorded in the database for processing alongside an acceptance message for the message broker system. The agent will utilize the light weight libraries to analyse page elements and extract contextualizing data for analysis and classification.

The chosen method for data collection is assisted through the utilization of two external API services to gather data alongside the direct processing of website data. The external

API service, SerpAPI, is implemented to scrape and extract data from search engine results and designed to provide a structured scope for searching the internet for social media scam content. The second API service is through BrightData, providing an external API to directly search social media platforms from sourced urls for social media content that may potentially be scam content.

```python
def build_mixed_queries(
    base_keywords,
    target_geography=None,
    target_organization=None,
    term_groups=None,
    patterns=None,
    max_queries=MAX_QUERIES
):
    core_query_parts = []
    if base_keywords:
        core_query_parts.append(base_keywords)
    if target_geography:
        core_query_parts.append(f'"{target_geography}"')
    if target_organization:
        core_query_parts.append(f'"{target_organization}"')
    final_core_query = " ".join(core_query_parts).strip()

    if patterns is None:
        patterns = [
            '" ({core_query}) AND ({terms})'
        ]

    if term_groups is None:
        term_groups = get_scam_lure_terms(base_keywords, target_geography, target_organization)
    term_groups = _expand_term_groups(term_groups)
    all_terms = []
    for group in term_groups:
        all_terms.extend(group)
    all_terms = list(dict.fromkeys(all_terms))
    queries = []
    mix_count = max(1, max_queries)
    for _ in range(mix_count):
        if len(all_terms) < 2:
            break
        mixed_terms = random.sample(all_terms, random.randint(2, min(4, len(all_terms))))
        terms_str = " OR ".join([f'"{term}"' for term in mixed_terms])
        pattern = random.choice(patterns)
        query = pattern.format(core_query=final_core_query, terms=terms_str)
        queries.append(query.strip())

    return list(dict.fromkeys(queries))[:max_queries]
```

Figure 4.3: SerpAPI Search Mechanism Implementation

Figure 4.3 illustrates the implemented search mechanism for SerpAPI and what serves each data collector and the anti-scam system the capability to search the internet through directed query searches. The implementation follows the modular design methodology of the data collection agents and requirements of the orchestrator agent to provide con-

trol variables used to direct the data collector agent. In the anti-scam multi-agent system the control variables "base_keywords", "target_geography", "target_organization", "term_groups", and "patterns", identified in the above figure are the first mechanism controls providing the orchestrator agent control tools. These control variables are what direct the data collection agent in its traversal of what sites to processes for scam related content on the internet and lead to the agents next key process, data processing.

Source Data Processing

The Data collection modules processing functionality comes from the combination of processing social content through the Brightdata API and direct page processing using the "GPT-5-mini" model with search functionality available via OpenAI. The implemented AI model for processing source data allows for the extraction of feature elements and item embeddings from dynamic loading pages. The item embeddings are generated from processing the provided response using the "text-embedding-3-small" model. The data collection agent is directed to use the "extraction_prompt" tool call to process any urls via the AI model and then directly make use of the subsequent functions; "parse_extracted_response" and "generate_embeddings" provided in the following figure:

```python
def parse_extraction_response(self, response: str, url: str) -> URLContentAnalysis:
    try:
        parsed = json.loads(response)
        entities_data = parsed.get('entities', {})
        entities = Entities(
            people=entities_data.get('people', []) if isinstance(entities_data.get('people'), list) else [],
            organizations=entities_data.get('organizations', []) if isinstance(entities_data.get('organizations'), list) else [],
            locations=entities_data.get('locations', []) if isinstance(entities_data.get('locations'), list) else [],
            products=entities_data.get('products', []) if isinstance(entities_data.get('products'), list) else []
        )

        return URLContentAnalysis(
            url=url,
            title=str(parsed.get('title', 'Unknown')),
            main_content=str(parsed.get('mainContent', '')),
            key_topics=parsed.get('keyTopics', []) if isinstance(parsed.get('keyTopics'), list) else [],
            content_type=self.validate_content_type(parsed.get('contentType')),
            entities=entities,
            sentiment=self.validate_sentiment(parsed.get('sentiment')),
            trustworthiness=max(0, min(1, float(parsed.get('trustworthiness', 0.5)))),
            reasoning=str(parsed.get('reasoning', 'No reasoning provided'))
        )
    except Exception as error:
        print(f'Error parsing extraction response: {error}')
        return self.get_default_analysis(url)

async def generate_embeddings(self, text: str) -> List[float]:
    try:
        response = self.openai.embeddings.create(
            model="text-embedding-3-small",
            input=text[:8000]
        )
        return response.data[0].embedding
    except Exception as error:
        print(f'Error generating embeddings: {error}')
        return []
```

Figure 4.4: Implemented Functions for Processing Sites and Extracting Embeddings

The final component of the data collection agent is the processing of data and extracting item embeddings to be used as apart of the classification process. Figure 4.4

demonstrates the implemented functions to achieve this goal and generate the preliminary inputs required for the classification agent to perform its operations. The functions are designed to be called via OpenAI tool calls but are identified as mandatory processes to which every piece of collected data will under go these progresses regardless of if results are produced. Social media content and related content with intent to scam can follow many different formats and for the anti-scam system to effectively detect and track evolving scams on social media and communication platforms, it must handle all forms of data regardless of if it contains specific elements. This design philosophy aims to assist in achieving the systems goals to collect and classify scam content of all sources and across multiple platforms.

4.3.2 Scam Classification and Verification Agent

The Scam Classification and Verification Agent is responsible for determining if data provided from the data collection agent is of a verified source, then classify the scam and provide a scam score representing the potential for the content to be scam material. The agent is powered by a LLM to leverage its ability for nuance natural language processing and understanding with the benefit of additional context. The classifier implements the use of the GPT-5 reasoning model via an API endpoints to perform classification tasks on records retrieved from the database. The agent will carefully filter the required data into the constructed prompts to provide the LLM with the information and instructions needed to complete operations and provide the scam classification alongside an evaluation on the context for a confidence rating. During the design process the decision to use a reasoning model was selected for the improved results when implemented with few-shot exemplars and RAG snippets to support LLM prompting. In comparison to intelligence models classification tasks using reasoning models proved to provide higher accuracy and lower hallucinations when dealing with large complex data evaluation tasks. The LLM result will provide the "scam type", "scam score" fields alongside the given data record to be collaborated and stored within the database. After classification and storage, an EVENT message will used as a data metric for the scam analysis agent and the orchestrator agent. The system will be designed for the orchestrator agent to act once the message is provided but the scam analyser agent will only act once a set EVENT message queuing metric is met.

In the pre-processing of data for the scam classification process, compiled data collections are generated using data retrieved from multiple tables in the database and serve as content records to be classified. The scam classification agent operates via a series of prompts each dedicated to assisting the model through contextualization and directing the final output. This is achieved through implementing the LangChain library into the AI agent and allowing multiple calls to be made to the LLM model using carefully constructed prompts designed to complete sub-tasks required to direct the designed output. Sub-tasks are crafted for each RAG implementation following similarly designed classification system methodologies using large language models as they identify RAG support for

prompting through exemplars as pivotal to successful classification of complex datasets. For this reason sub tasks that provide the LLM with supporting classification exemplars directed by the orchestrator agent are also individually crafted prompts with supporting context to also direct how example material should be used for the classification process.

```
// Main function to perform LLM-based scam classification assessment
// core analysis function for scam classiciation agent to produce scam assessment and scam score
async function llmClassificationAssessment(
    dataRecord: DataRecord,
    contentInfo: ContentInfo
): Promise<LLMAssessment> {
    try {
        const promptTemplate = buildPromptTemplate();
        const contentSection = buildContentSection(dataRecord, contentInfo);
        const formattedPrompt = formatPrompt(
            promptTemplate,
            getSystemContext(),
            getAnalysisInstructions(),
            getRedFlagIndicators(),
            getClassificationGuidelines(),
            getOutputFormatInstructions(),
            JSON.stringify(dataRecord, null, 2),
            contentSection
        );
        const response = await LLM.invoke(formattedPrompt);
        const parsed = contentParser.parse(response.content);

        const embeddings = generateEmbeddings(
            `${contentInfo.title || ''} ${contentInfo.body || ''} ${contentInfo.description || ''}`
        );

        return {
            recordId: dataRecord.id,
            scamScore: calculateScamScore(parsed.trustworthiness, parsed.contentType),
            confidence: parsed.trustworthiness,
            scamType: determineScamType(parsed.contentType, parsed.keyTopics),
            assessment: mapToAssessment(parsed.contentType),
            reasoning: parsed.reasoning,
            redFlags: extractRedFlags(parsed.keyTopics, parsed.reasoning),
            entities: parsed.entities,
            embeddings
        };

    } catch (error) {
        console.error('Error in llmAssessment:', error);
        return getDefaultLLMAssessment(dataRecord.id);
    }
}
```

Figure 4.5: Implemented Classification Function from Scam Classification Agent

Figure 4.5 provides the implementation of the core scam classification function and overall goal of the scam classification and verification agent. The external control vari-

ables used by the orchestrator agent to control agent functionality within limitations are implemented in the "getClassificationGuidelines()" and "getOutputFormatInstructions()" prompt functions. Both functions implement the orchestrators ability to control system operations and adjust agent operations, within limitations, to influence agent operations. The scam classification agent contains two post endpoints, one to control RAG exemplar selection from the share knowledge base and the other to receive a JSON with classification label outlines and scoring criteria. The control limitations are limited to adjusting the few-shot limitations of the classification agent and specific labels to select from when retrieving from the knowledge base and processing model results. The final model returning results from the subsequent LLM calls are processed to categorize the assessment, produce suitable labeling for fields required in scam trend analysis, and assist when comparing clustered results or trends.

4.3.3 Scam Trend Analysis Agent

The scam Trend analysis agent will operate on the system knowledge base at a given interval depending on the amount of data collected to provide a context window either controlled via the RabbitMQ system or orchestrator agent. The trend analysis is confuted using a large language model for the analysis of results and the data correlations are achieved through a byzantine analysis procedure and item bedding evaluation pipeline. The trend analysis will be conducted using a large language model and for this instance the Claude model sonnet will be used; "Claude-3-7-sonnet-20250219". The use of this model is required for the following operation of the agent due to its ability to process larger amount of data in comparison to other models and ability to draw conclusions from recurring prompt submissions as it is able to handle entry input in dedicated file formats in conjunction with select prompts and then the model will cycle through prompts alongside entries. This possible due to its ability to process 120,000 input tokens without turn limiting its resulting output token limit and by setting a temp metric the scam trend analysis agent is able to process a different number of groups every time with the same level of accuracy. The output of the scam trend analysis agent will be a series of report statistics individually tailored to the specific type of scams in comparison to previously concluded results.

```
//Analyzes temporal patterns, topic trends, and anomalies in the dataset
async function trendAnalysis(dataSet: DataSet): Promise<TrendAnalysisResult> {
    try {
        const timeSeriesData = extractTimeSeriesData(dataSet.records);
        const topicTrends = identifyTopicTrends(dataSet.records);
        const anomalies = detectAnomalies(timeSeriesData, dataSet.records);
        const summary = generateTrendSummary(topicTrends, anomalies);

        return {
            trends: topicTrends,
            timeSeriesData,
            anomalies,
            summary
        };
    } catch (error) {
        console.error('Error in trend analysis:', error);
        return getDefaultTrendAnalysis();
    }
}
```

Figure 4.6: Implemented Classification Function from Scam Classification Agent

This collection is stored within the database separately to the raw data records with appended types for progressive orchestrator analysis. In the process of trend analysis the pandas library is also be integrated into the internal pipeline of the agent for data pre-processing although this is to provide the large language model with comparative result to its output as apart of an internal self learning process as well as accuracy indicator. If the constant resulting data grows in difference the orchestrator agent will take immediate action to adjust accuracy levels. This is also used to ensure that a quantitative measure can be correlated to a contextual output for the evaluation of the Trend analysis agent by the orchestrator agent. In figure 4.6 we can see the overall implementation of the trend analysis process implemented through the 4 main processes. The return data of each process is what is provided to the overruling LLM assessment and are also provided to the orchestrator agent through the RabbitMQ message queue.

4.3.4 Orchestrator Agent

The Orchestrator agent is the operational director that implements self evaluation and adaptation to evolving scam tactics employed by cyber criminals to continuously evade non analytic platform moderation. Under its direction within the system architecture are three vital large language model powered agents, the web scraper agent that will interface with social media platforms, the scam classifier agent to process and enrich data, and the scam trend analysis agent to aggregate information over time.

The use of a single centralized database serves as the storage facility for the results

generated from these agents. Communication between the system agents is asynchronous via message queues allowing individual agents to operate independently of one another and the communication pipeline operates using five self-learning cycles. The architecture loosely modeled after a microservice design where each agent has a designated responsibility and interfaces with others for collaboration. Although updates are notified through the message queue system implemented using RabbitMQ server on the Kubernetes cluster the individual endpoint calls are what change configurations.

The ochestrator agent is the overruling authority of the multi-agent system and will employ a number of comparative conclusion methods to measure the performance of individual agents and direct the overall system based on a collective assessment of the combined results. The agents first key feature is the sequencing and scheduling of the other agents to maintain a consistent flow of communication through the message broker framework. The agent will provide data validation and schema checks for the collective database as an additional health check measure for the initial stages of the pipeline using. This process will be conducted similarly to the scam trend analysis agent but this specific task along within the supervisor agent will use the "Claude-3-7-sonnet-20250219" model for the ability to single prompt a collective entry list and let the LLM handle the individual generation of single input prompts, limiting the number of tokens used in total during data pre-processing as this process is not considered critical because the cleaning of data could effect the learning process cycles employed throughout the system. The orchestrator specific responsibilities in regards to each individual agent themselves are:

- Directing the Web Scraper Agent through adjusting the light-weight LLM controller prompting search queries.
- Monitoring database and event message counts to notify Scam Trend Analyser if it needs to conduct an evaluation on the data before its preset time interval is triggered.
- Analyse outputs from both the scam trend analyser agent and scam classifier agent to evaluate their performance and accuracy against pre-set quantitative measures.
- Fault tolerance of agents via its integrated ability to take on tasks if there is component failure (Will be further explained in the following section).
- Failure stability of agents via task and operation freezing if any external sources are not available. (e.g. if OpenAI or Anthropic platforms cannot be reached)
- RabbitMQ record keeping by correlating Developer, Consumer, Queue.Name(Action)

The orchestrator agents decision making logic is implemented as a combination of both algorithmic based imposed rule sets and LLM data evaluation results. The standalone LLM used by the orchestrator agent will be prompted an appended list of quantitative measurables (queue size, operating run-time, rate limitations from external sources on

agents, amount of data collected, speed of data extraction) collected from the other agents and the resulting outputs will be compared through a deterministic algorithm to adjust internal variables. The adjusted variables are those used by the supervisor agent to control the operative authority over the other agents within the multi AI-agent system. The self adjusted monitoring metrics and control variables are encapsulated in preset upper and lower boundaries to maintain control flow determinism and predictability within the system while allowing adaptability. This internal self adjustment is the core of the systems adaptive learning process needed to maintain an evolving solution to match the increasing rate of change in scam activity on social media platforms. LLM combined integrated logic for flexible internal rule adjustments is a revolutionary solution to system self learning through the ability to internally adjust its measurables used in the adjustment and directing of other AI-Agents within the system. Through this implementation above the web scraper agent, scam classifier agent, and scam trend agent can operate without needing to use hard codded logic and coordinate as a cohesive system.

4.4 System Technologies

Technology	**Role and Rationale**
Selenium and Playwright Combo	Used for the web scraper agent to automate browser control for data collection. Able to handle dynamic JavaScript execution and endless feeds. Bypass social media platform security measures. Reason for choice: ability to handle complex HTML while still executing dynamic JavaScript loading.
BeautifulSoup	Parses the provided DOM and extracts required data for system components. Reason for choice: easier to use when working with when dealing with dynamic field names.
OpenAI API (gpt-4o)	Used for Scam Classifier Agent to access GPT-4o and its advanced reasoning models as these are benchmarked to be considerably better than other LLMs for precise classification. Reason for choice: reliability and excellent performance for complex multi steps tasks that require reasoning
Anthropic API (Claude-3-7-sonnet-2025XXXX)	Used for Scam Trend Analysis Agent and Supervisor Agent, allows for large amounts of data to be processed with a high level of accuracy. Reason for Choice: reliability and efficiency when dealing with large datasets at considerable speeds without hindering accuracy

Technology	**Role and Rationale**
Embeddings and RAG (Text-Embeddings-3-Large)	Used in the Scam Trend Analysis Agent for similarity searching, duplication removal, and entity grouping. Enables the ability to normalize content indexes to be able to search k-NN across recent and historical data. Reason for choice: Reduces hallucinations while improving recall for operation in evolving scam environments. Provides quality dense representations with cross-domain performance and scaling in vector indexes.
Hierarchical Density-Based Spatial Clustering Applications with Noise (HDBSCAN)	Used by the Scam Trend Analysis Agent to discover dense clusters of related scam classifications without needing to provide specified cluster information. Reason for choice: Able to handle variable density data while labeling outliers as noise and provides cluster scoring that translates to a measure of confidence.
K-Mean / Mini-BatchKMeans (MBKM)	Used by the Orchestrator and Scam Trend Analysis Agents to process large batchs of trend report data. Supports partial fitting of vectors and enables re-clustering without extensive recomputation of data. Reason for choice: allows agents to easily interpret data for analysis reports and supports agent planing stages with deterministic outcomes.
RabbitMQ	Message broker for asynchronous communication between agents. Reason for choice: decouples agents in relation to type and provides complex routing support and fault tolerance in multi agent systems.
Prisma ORM and PostgreSQL/AmazonRDS	Object Relational Mapping Storage solution. Relational database and provides JSON support querying. Reason for choice: ease of use and ability to support structured relationships.
Pandas	Used for comparative evaluation of LLM when generating Scam Trend reports. Additionally used for quantitative measurables, accuracy, precision, and recall. Reason for choice: ability to integrate Scikit-Learn in future works and development.
API Implementations	RESTful API to make quick adjustments to overall system via the orchestrator agent. This implementation was selected to alleviate message queuing issues and separate system control functionalities from the system control components.
Amazon Elastic Kubernetes Service	running environment for all agents in the system with additional deployment and update support. Additionally provides secret management and dynamic environment variable controlling for changing configurations. Reason for choice: Integrates seamlessly with Amazon Web Services for cloud hosting and leverages AWS networking for system operations.

Technology	Role and Rationale
Amazon Web Services	Cloud platform for hosting the proposed anti-scam system. Provides a variety of tools such as AmazonRDS(PostgreSQL database service) and AEKS with pre-configured hosting services. Reason for choice: seamless integration with other hosting services needed to run the proposed system and provides system metric records for testing purposes.

4.5 Artificial Intelligence Models and Prompt Engineering

4.5.1 Prompt Engineering

The multi AI-Agent scam detection system for social media makes use of multiple large language models, these being GPT from OpenAI and Claude from Anthropic. Prompt Engineering is a critical component of utilizing LLMs effectively and the output of these models is deeply relent on the quality of the prompts wording, formatting, and context that is provided alongside with it. Each Agent in the scam detection system has specifically curated prompts as well as being provided an output structure to follow for its generation to minimize any uncertainty and unpredictability. Several techniques are employed to create these prompts, the first being that at the start of each prompt, the LLMs role is clearly outlined to provide it with foundational context to the task. For the scam classifier agent the second step is to provide a set of classifications for the LLM to choose from for its conclusion on the given input. This is the prompt engineering technique most commonly used for classification tasks where constraints are placed on the potential outputs to significantly decrease randomization and eliminate unwanted data generation. The second step in the prompt engineering process for the scam trend analysis agent is to provide the LLM with prioritization for which data is to be considered more important and the order in which these considerations must take place. In the case of both the scam classifier agent and scam tend analyzer agent, the next part of the prompt is to provide the LLM with a carefully constructed list of examples to provide additional understanding to what components from the input data generate accurate outputs.

Once the standardized prompt template has been created for both the scam classifier and trend analyser AI-Agents, a series of refinement through testing is conducted. This will be achieved my passing a series of manually generated data with pre-determined outputs to the prompt, adjusting generation parameters or including chain of thought additions. Adjusting generations hyper-parameters is more effective for classification tasks like that of the scam classifier agent, while chain of thought additions to prompts see considerably more effective results in analysis tasks where inferences in the data can be harder to replicate. In either case, this process of iterative refinement and evaluation is the core to prompt engineering, Refer to Appendix A for a demonstration of iterative

improvements on the early stage prompting for the scam classifier agent. For both the scam classifier and trend analyser agents repeated iterations also helps the specific model internalize the expected output allowing the model to begin being able to infer the pattern from these demonstrations and apply it to new instances moving forward. The conceptual prompt engineering framework, ReAct, is an internal variation of what is going to be used to develop specifically tailored prompts for the individual AI agents in the scam detection system for social media content. ReAct refers to Reasoning and Action prompting where your prompt the model to reason about the problem, then choose an action, and iterate through that cycle a select amount of times to evaluate comparative accuracy. For the proposed Multi-AI Agent Scam Detection System

In this Thesis all prompting will follow an iterative approach to curating and refining the initial prompts which will then be enhanced using the LangChain framework. This framework was selected as it will allow the specific agent LLMs in the system to keep record of earlier messages and utilize the tables within the database as a continuously growling list of examples to assist in their dedicated tasks. The use of this framework is also why RabbitMQ was selected as the systems dedicated message broker. The combination of database connectivity and the storing of the fluctuating communication pipeline in the proposed multi AI-Agent system will assist in the supervisor agents self learning process, providing it with memory regarding specific system operations that are critical to the Supervisor agents decision making process.

Chapter 5

System Testing Methodology

The following chapter outlines the methodology implemented for the evaluation of the proposed closed loop scam detection and classification system as well as individualized model evaluations. The testing process is designed to comprehensibly test individual modules and compare their performances in both the open loop and closed loop architectures. The two architectures will be compared through the evaluation of the specific module components and then overall system outputs following a black box testing methodology to see if internalized feedback loop for self learning produce continuous improvement in AI-Agent systems. Manual testing procedures were used for the individual testing of each agent following a gray-box testing methodology to incorporate internal model results into the comparison of final results. Module testing happened both on an per architecture basis and individual implementation testing for core functionalities. Architectural level performance metrics where also compared through deployment infrastructure tools provided by the services to maintain fair system comparisons during the live testing phases.

5.1 Validation Objectives

The primary aim of the system architecture validation process and individual module validation processes is to comprehensively evaluate the effectiveness in achieving a system that is capable of evolving alongside the scam environment and operate in real time to detect and track scam content on social media and communication platforms. The key objectives the validation process aims to achieve are:

- Evaluate the effectiveness of internalized self learning loops in a system architecture and its ability to progressively adapt to its environment and changing input data.
- Evaluate the effectiveness of an additionally implemented orchestrator agent to control module operations through specialized control variables and tool calls.
- Access the collection rates and number of newly collected pieces of data over time to stagnant data collection agent search inputs.

- Evaluate the effectiveness of Large Language Models for classification based AI-Agents for complex multi-process tasks.
- Compare architectures implementations and evaluate the impact of architecture type of system operations over time.

In order to achieve these objectives, the testing process will be conducted following three testing methodology implementations, the first being the individual testing of each modules core functionality to produce realistic acceptance criteria through gray-box testing. The second method is individual testing of agent functionalities in respects to the architecture implemented. Then finally the overall system testing of both architectures through black-box testing to compare final results over time. The black box testing process will also implement a live testing stage to compare results from real work implementations and provide insight to assist in evaluating the two architectures being compared.

5.2 Testing Environment

All testing procedures conducted followed standardized testing configurations outlined as follows; All architecture and system level testing occurred in isolated through separate Kubernetes name spaces per architecture and identical agent deployments. Additionally all testing was conducted using strict version controls for all external API services, AI models, Large Language Models, and embedding versions. The testing components that come along with each AI implementation follows the same token count and temperature limitations applicable to the model in use. The final testing configuration is conduced on a per module basis and uses the same selected dataset for all module function testing to implement a consistent testing environment and comparison of results.

Labeling of testing records was achieved through the live development and generation of the testing dataset. Links where collected from various OSINT libraries and passed to the data collection agent during individual module testing to collect the required information for pre-defined scam classifications. In addition baseline system cycles where conducted to create testing datasets and performance based benchmarks for progressive functionalities.

5.3 Agent Testing Methodologies

Four key testing metrics where used for the designed module testing methodology, being able to produce numerical results to evaluate classification and analysis processes specifically. The breakdown of testing metrics is as follows, the ROC-AUC metric will be used to evaluate the classifications scam score component and identify the ability to prioritize positives over negatives for scam category outliers. The F1 score for the agents ML decision layer will provide a numerical representation of module operation quality at the point of reference. The accuracy will support the assessment of classification and analysis

processes by providing overall correctness but will only server as a secondary measurement due to variable unbalances and potential to hide skewed results through poor recall. Finally the integrated self learning loops will be tested through progressive RAG results and the precision with relevant retrieval examples(precision at K).

- **ROC-AUC (Receiver Operating Characteristic Area Under the Curve):** provides the probability a random positive has a higher scam score than a random negative. Both the positive and negatives are labeled through scam classification. The items are processed and sorted by scam score descending where then the integrated section of the precision *(Precision = true positive / (true positive + false positive))* vs recall *(Recall= false positive / (false positive + true negative))*. Modular testing metric will be used for the scam classification agent.

- **F1 - Precision and Recall:** giving the numerical metric for comparison of single point operation quality for the current system configuration. Adapting configurations F1 score will be compared across time windows to evaluate progressive system functionality in the closed loop architecture. *(Given as F1 = (2.Precision.Recall) / (Precision + Recall))*

- **Accuracy:** is only measured for final architecture analysis results given as the correctness of all predictions when using progressing datasets. The datasets are generated using data already classified as a potential scam proving both detrimental to the classification of true negatives but no benefit to the classification of true positives. This payoff is why the accuracy of the system architectures is included to support comparison results.

- **Top K Precision:** selected as the evaluating metric for the RAG implementations and system self learning loops via the orchestrator and classification agents. Overall this will provide insight if the given retrieved items are actually relevant to the query and assisted in the LLM producing an accurate assessment. This specific test is what RAG exemplar from the selected dataset constructed through manual intervention helps the model classify the scam content as well as produce an assessment to explain the scam classification and score.

To finalize the specific agent testing procedural relations for testing metrics and methodologies, the data collection agent will be tested following a black box testing methodology to gather results regarding the collection rate and number of new items found over a time frame. Both testing metrics will provide a solid foundation for the performance evaluations of the data collection agent in both architectures and as a stand alone agent operating over time. This performance evaluation will also serve to support the use of a combined web scraper and crawler library design and external API collection rates. The scam classification agent will be tested to gather result quality and accuracy over time for the classification and scoring process of items in the system. These results will server to evaluate agent performance inclusion of self learning loops in the differing system architecture implementations. The scam trend analysis results will be gathered to

evaluate the classification of resulting clusters and compare the trend assessment results impact on orchestrator agent large language model outputs.

The orchestrator agent testing process will be conducted through manual testing components and functions at the operation level then by evaluating the influence of the agent in the closed loop architecture. The open loop architecture results will serve as the baseline target results for the orchestrators influence on the closed loop architecture and the resulting difference will present weather improvements are made. Alongside this two individual implementation testing results will be evaluated for the individual agent performance testing, these being time to action lag and idem-potency success proving that the systems configuration are just guidelines and not restrictions.

5.4 Architecture Testing Methodology

The architecture testing methodology aims to provide a means to evaluate the implementation of internalized self learning loops in a system architecture when compared to another without feedback. To evaluate the performance of the open loop and closed loop architectures a series of controlled experiment operations were conducted under identical environment conditions. The black box testing method was followed for live testing procedures in order to collect final system architecture results. This testing process will allow the assessment of how architecture structure influences key system functionality leading to system evolution while still operating in the same environment. The closed loop architectures configuration will change over time through the orchestrator agents operations but these evolving configurations will be arrogated in the final architecture comparison as only the process and not the specific alterations are being evaluated.

Metric	**Type**	**What it provides**
ROC–AUC	Ranking (no τ)	Ranking quality under imbalance.
F1@τ	Operating point	Balance of precision/recall at a point.
Precision	Operating point	Cleanliness of classifications.
Recall	Operating point	Catch rate of scams
Specificity	Operating point	True negative rate; false-alarm control.
Accuracy	Operating point	Overall correctness; secondary under class imbalance.
Precision@K	Top-K / ranking	Quality of top items/evidence; predicts downstream gains.

Table 5.1: Compact metrics used to compare the Scam Classification Agent. Use PR–AUC (ranking) and F1@τ (operating point) as primaries.

The architectural testing phase for functional metric collection is achieved through the completion of the agent testing procedures and agent integration testing results. The performance evaluation is achieved through the overall testing of the architecture in a testing phase and compare record statistics for processing time and record collection rates. In both testing phases the architecture agents where verified to ensure all operations work correctly and initial test scripts were run to validate the production of equivalent logical

outputs for identical inputs. Qualitative testing was conducted during the performance testing phase through observing system adaptability and fault tolerance to varied datasets. The closed loop systems self learning mechanism was the testing focus for intentionally introducing variations and comparing results against the static behavior of the open loop pipeline over the same data inputs.

Chapter 6

Results

Results section as follows: Open loop architecture baseline results agent evaluations and testing results closed loop architecture testing results and comparisons

6.1 Baseline Results

The open loop architecture testing process evaluated the baseline architecture and module implementations under the outlined testing configuration for ten cycles. The average resulting metric from each test run over each cycle is used as the target goal for subsequent closed loop architecture and agent testing processes. These results used the same configuration setup for all agents in the system, while following tests will use the same configuration in the initial cycle, system functions adjust these through every iteration or after multiple cycle completions. All results in table 6.1 are an average of the final results run on the open loop system over 5 test runs of 10 cycles.

Module / Agent	**Metric**	**Result (Target)**	**Pass?**	**Notes**
Data Collection Agent	Recall	0.89 (≥ 0.85)	Yes	Compare to previous cycle results
	Discovery Success Rate	avg 12/min (≥ 10/min)	Yes	Observation $\rightarrow$ successful process
Classifier Agent	ROC–AUC	0.84 (≥ 0.80)	Yes	Ranking on test dataset
	F1@τ	0.78 (≥ 0.75)	Yes	τ fixed from dev; confusion matrix verified
Trend Analysis Agent	Cluster labeling	0.89 (≥ 0.85)	Yes	Analyst-verified trend alerts
	Trend detection lag	3 cycles (≤ 5 cycles)	Yes	New pattern surfaced to analysts

Table 6.1: Open Loop Agent Test Results (Initial).

The results displayed in Table 6.1 provide an overview of initial agent specific testing results for the open loop agent architecture. The data collection agents recall achieved 89% although the testing dataset used was consistent on every cycle and the recall can display a skewed results as less overall items are ingested into the system pipeline. The classifier agent operated with a single call knowledge base for the RAG exemplars therefore baseline agent data is averaged as an overall of all testing cycles to remove the effect that the datasets processing order has on the overall agent results.

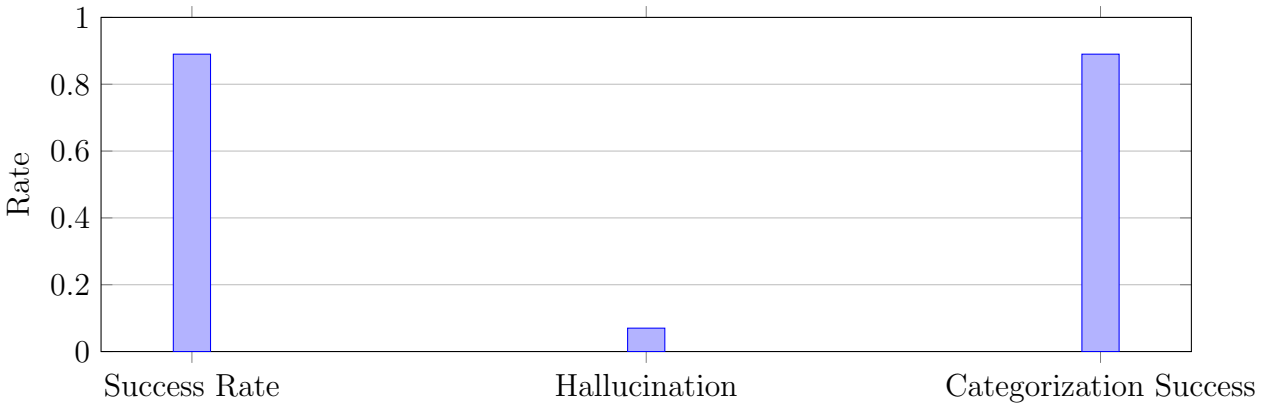


Figure 6.1: System processing successfully with one item extracted from source (Open Loop Architecture)

Figure 6.1 indicates the success rate of a system processing a source and extracting an item, with the dataset containing no true negatives, the results demonstrate a high achievement rate and expected outcomes from initial cycles of the closed loop architecture. The testing results in this figure carry on to be the comparative baseline because initial cycles of the closed loop architecture use the same configuration. The results for success rate and categorization success should see influence from the inclusion of the orchestrator agent in architecture comparisons although the average hallucination indicating 0.07 this should stay the same or increase slightly in all other system testing procedures. The models operating at the same token and temperature limitations following the testing methodology description will show the same results unless there is a change in agent specific calling which for testing purposes all agents followed the same development version.

6.2 Agent Testing Results

Data Collection Agent

The results collected from the data collection agent testing process are gathered in the table as the overall average of all results after a complete testing phase, a phase indicates that the system operated through 10 cycles successfully. The open loop architecture displays a discovery success rate of 12 as the average cumulated data from every cycle and subsequent cycles in the closed loop architecture produced fewer results.

Metric	Open Loop	Closed Loop
Recall	0.90	0.93
Discovery success rate	12	25
Process success rate	0.95	0.97
Retry success rate	0.80	0.88
Initial Throughput (items/min)	1800	1650
Throughput (items/min)	1600	1900
Unique URL ratio (distinct/source)	0.88	0.92

Table 6.2: Data Collection agent comparison: Open Loop vs closed loop architecture.

The irregularity in Table 6.2 is further highlighted with the initial throughput of the system architectures favoring the open loop architecture due its lower time to process an item which then balances out as the overall number of items discovered in subsequent cycles decrease in the open loop architecture. With a decrease from 1800 to 1600 comparative to the increase in throughput for the closed loop architecture the changing search configuration demonstrates an impact from the internalized self learning and adjustment loops.

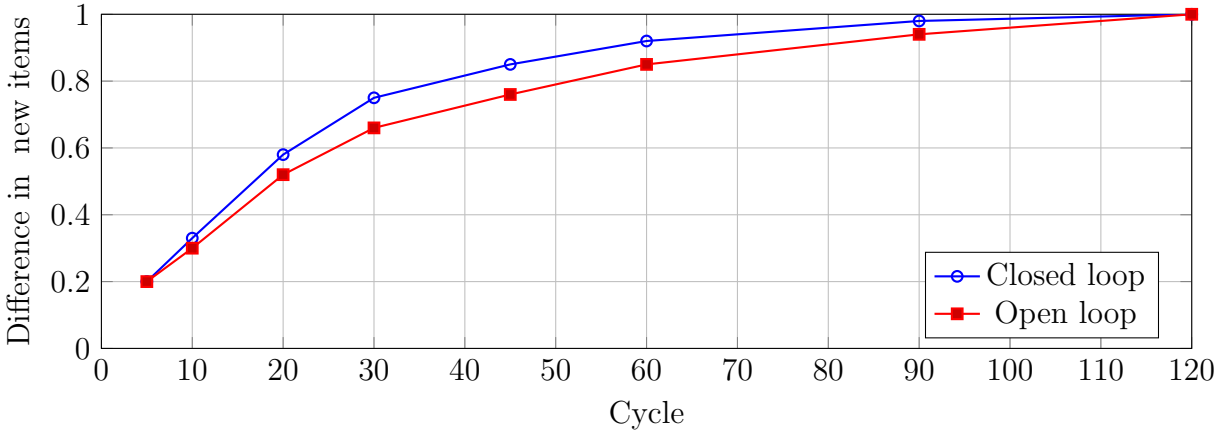


Figure 6.2: Crawler new discovery curve.

The data collection Agent was also evaluated through the difference in the number of new items collected throughout the data collection process. The evaluation will show how the effect on internalized self learning loops in the closed loop architecture were able to control configurations for the data collection agent to adjust controls and improve performance.

Scam Classification Agent

The scam classification agent results demonstrate the final results of the scam classification agent in both architecture types following a set of 8 program parses. These parses

where conducted under the same configuration settings as the data collection agent followed the same 10 cycles per parse for the final average results to be produced. The systems classification was analysed from these results and through a comparison the implementation of the scam classification agent in each architecture can be concluded. Table

Metric	**Open Loop**	**Closed Loop**
Precision	0.89	0.93
Recall	0.88	0.94
F1 score	0.89	0.94
ROC–AUC	0.92	0.96
Accuracy (analysis only)	0.95	0.96
False Positive Rate	0.7	0.07
False Negative Rate	0.8	0.06

Table 6.3: Scam Classification agent comparison: Open Loop vs Closed Loop architecture.

6.3 presents the final findings of each architecture type for the final comparison. We can see that the false positive rate and false negative rate in both classification comparisons doesn't show any improvement over the entire testing process. This can be concluded from the overall number of classifications that the closed loop architecture made as the system previously demonstrated that it had a higher successful processing rate. Although the rates indicate no improvement in classification for failure rates, we can visualize the improvement in over system operations through the close loop architecture. The level of improvement for the specific metric indicates the value in implemented self learning loops and feedback into the social media and communication anti scam system.

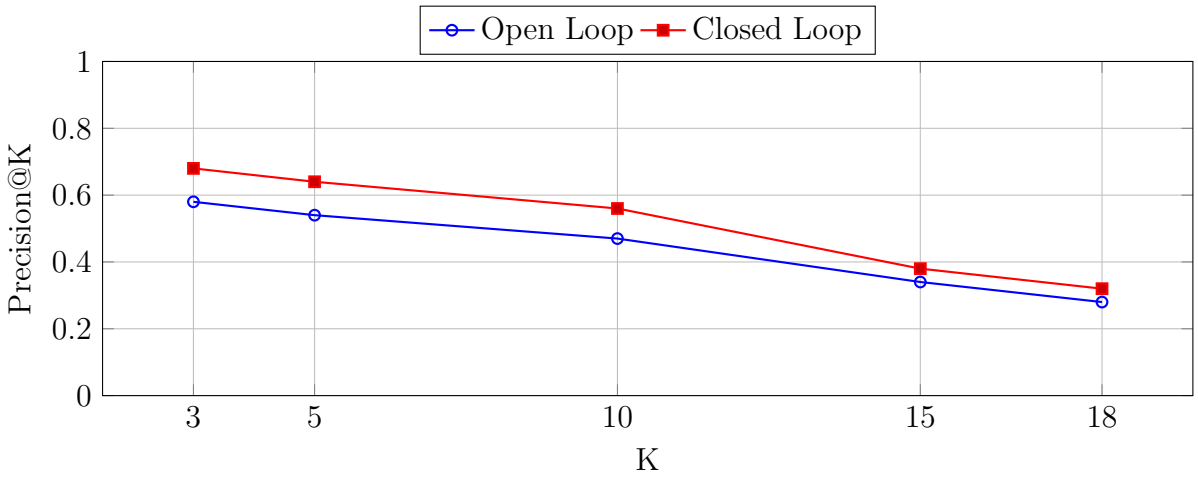


Figure 6.3: Retriever Precision@K.

In the evaluation of the implemented self learning control loop through the orchestrator agents control of the scam classification agent RAG selection pool, the final resulting tests

displayed in Figure 6.3 show us that as K increase so too does the overall probability of relevance of the exemplar content. An indication that the decreased accuracy for the retrieval component is consistent for K is good for overall system evaluations but shows that in a closed loop architecture, providing additional LLM support must be specifically limited to evidence that is beneficial to the outcome.

Scam Trend Analysis Agent

The scam trend analysis results are analysed for both the identification of trends and similar characteristics in similar data clusters. The Analysis agents overall performance in the case of the system goal is the resulting impact its performance and output has on the system performance overall and the impact an outcome has on further system decisions.

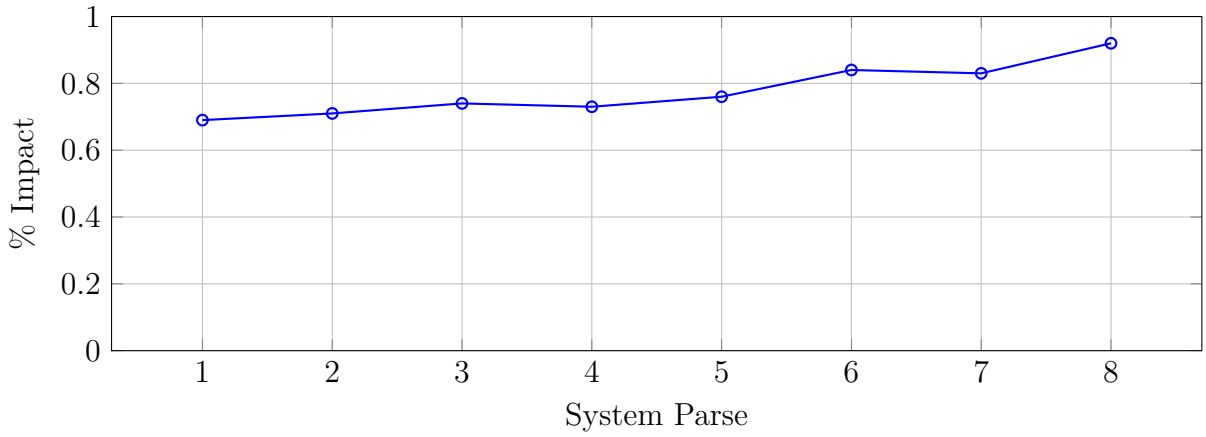


Figure 6.4: Trend clusters Impact on System through Orchestrator agent

Metric	Open Loop	Closed Loop
Precision@K	0.82	0.89
ROC–AUC (trend detection)	0.78	0.86
Topic Labeling	0.34	0.47
Pairwise dedup F1	0.76	0.84
Forecast failure	18%	12%
Drift detection recall	0.71	0.86
Item process Rate	540	620
Deduplication rate	0.73	0.86
Index refresh time	90	60

Table 6.4: Scam Trend Analysis agent comparison: Open Loop vs Closed Loop architecture.

Table 6.4 is the Scam Trend Analysis Agents final results for agent implementation comparison for both architecture types as well as the system evaluation to identify if

internalized self learning loops can allow a system to adapt to an evolving environment. Final results for the scam trend analysis displayed table 6.4 where conducted following the gray box testing methodology outlined in the previous chapter and can be compared to the progressive impact each analysis had on system operations through the orchestrator agent in figure 6.4. This comparison allows for the analytical evaluation of the scam trend analysis and internal system feedback loops function in the closed loop architecture without human evaluation of results. The comparison also shows that the final results for the scam trend analysis are only a snapshot of system operation but with the visual representation of differing results through each parse, we can see the impact progression over time.

Orchestrator Agent Agent

The final evaluation of agent operations is the internal operations of the orchestrator agent in the closed loop architecture. They key mechanism the orchestrators functionality applies is the adjustment of other agent operations in the system, figure 6.5 demonstrates the number of calls made at each cycle in a parse as an average of all collected data. This figure demonstrates the high disparity in the average number of calls at each cycle and potential effects that lead to the following dropoff. This peek is encountered at the midway point of the testing process and alone only gives insight into orchestrator performance but when compared to results from the other agents we can drive the impact the number of calls has to the overall change in system operations and resulting system outputs in the closed loop architecture.

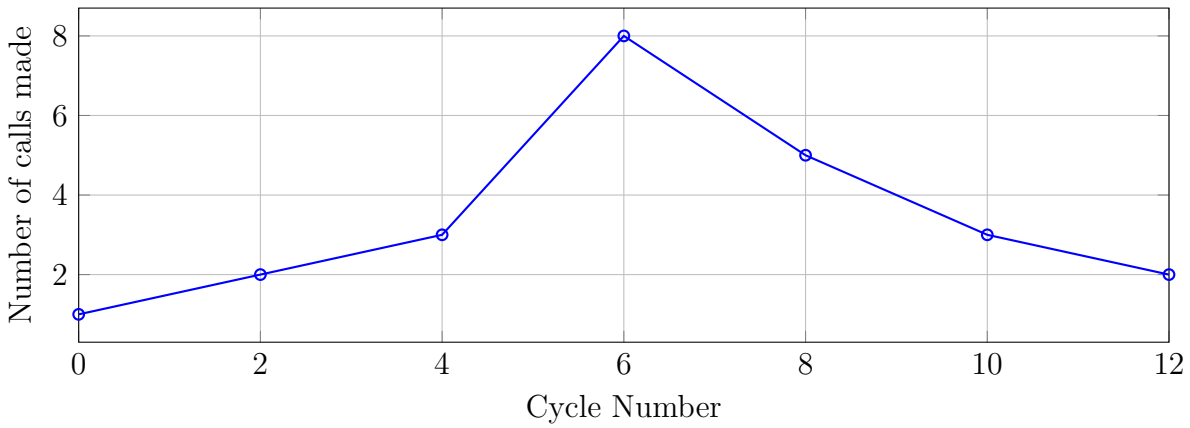


Figure 6.5: Orchestrator number of calls made to Queue adjustments.

Each time that a call is made we can see the rate at which each agent was called per cycle and determine the effectiveness each call and the number of calls has on the overall system output when comparing results over the testing process.

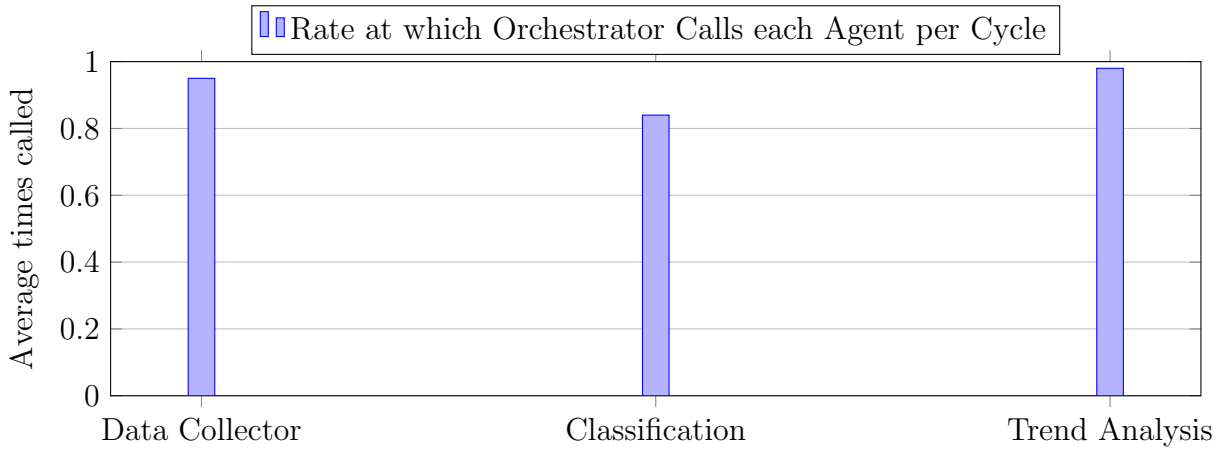


Figure 6.6: Rate at which Orchestrator Calls each Agent per Cycle

Figure 6.6 was collected from the RabbitMQ knowledge base to serve as the metric to evaluate the orchestrators functional test of directing agent communication through each endpoint. A change in system operations indicates a successful impact from the call and each changes impact is evaluated by the overall system effect weather positive or negative.

6.3 Closed Loop Architecture Results

The final testing phase was conducted for the overall testing of the closed loop architecture and evaluation of the impact that internal self learning loops and feedback have an a multi agent scam detection and classification system for evolving social media and communication environments. A general agent metric overview is provided in table 6.5 indicating all the key metrics that passed the baseline architecture results.

Module / Agent	**Metric**	**Result (Target)**	**Pass?**	**Notes**
Data Collection Agent	Recall	0.92 ($\geq$ 0.89)	Yes	Compared against baseline process results
	Discovery Success Rate	avg 25/min ($\geq$ 10/min)	Yes	Observation $\rightarrow$ successful process
Classifier Agent	ROC–AUC	0.92 ($\geq$ 0.84)	Yes	Ranking on test dataset
	F1@τ	0.84 ($\geq$ 0.78)	Yes	τ fixed from config; confusion matrix verified
Trend Analysis Agent	Cluster labeling	0.98 ($\geq$ 0.85)	Yes	Analyst-verified trend alerts
	Trend detection lag	2 cycles ($\leq$ 3 cycles)	Yes	New pattern surfaced to analysts
Orchestrator Agent	Action lag	5 min ($\leq$ 5 min)	Yes	Time between final results and adjustments
	Idempotency success	99.95% ($\geq$ 99.9%)	Yes	Duplicate config doesn't create duplicate outputs

Table 6.5: Key module/agent test results (compact layout).

In this table we can view the results when compared to the previous open loop overview table in figure 6.1. The analysis was conducted to achieve results greater or less than the expected results provided from the previous testing iterations of the open loop architecture. Indicating a pass if the metrics achieved expected results is how the analysis of each agents achievements impact and are impacted by changes in system operations.

Evaluating the closed loop system architectures agent components to conclude if internalized self learning loops impact agent operations is displayed through a series of graphs. Each graph displays the visual results for the comparison of both architecture types agents components in a specific task. Following the structured stages of the testing methodology the final results from each agent component is what will provide evidence for conclusions derived from the project work.Each of the provided graphs is used to demonstrate the individual mechanism metric result to be analysed and conclude findings in the discussion.

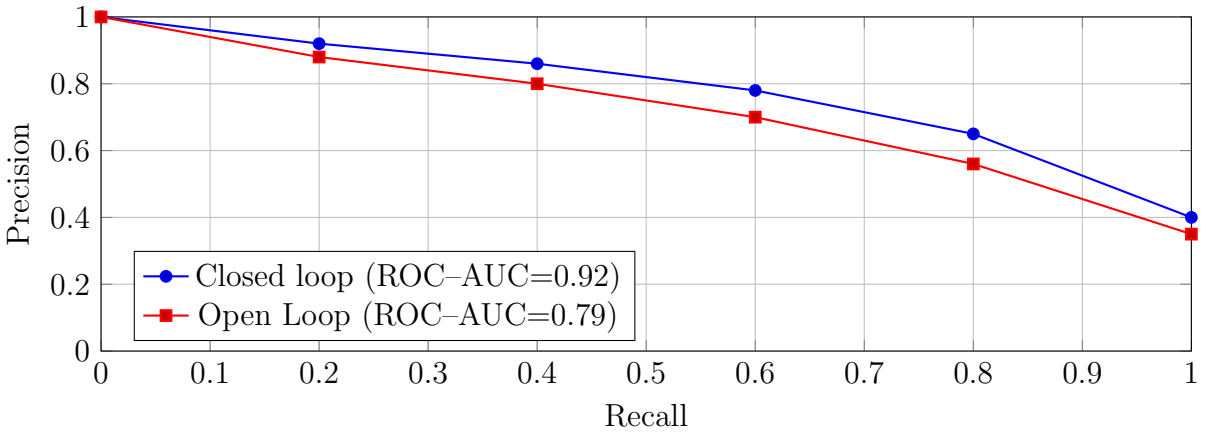


Figure 6.7: Classifier Precision–Recall curve.

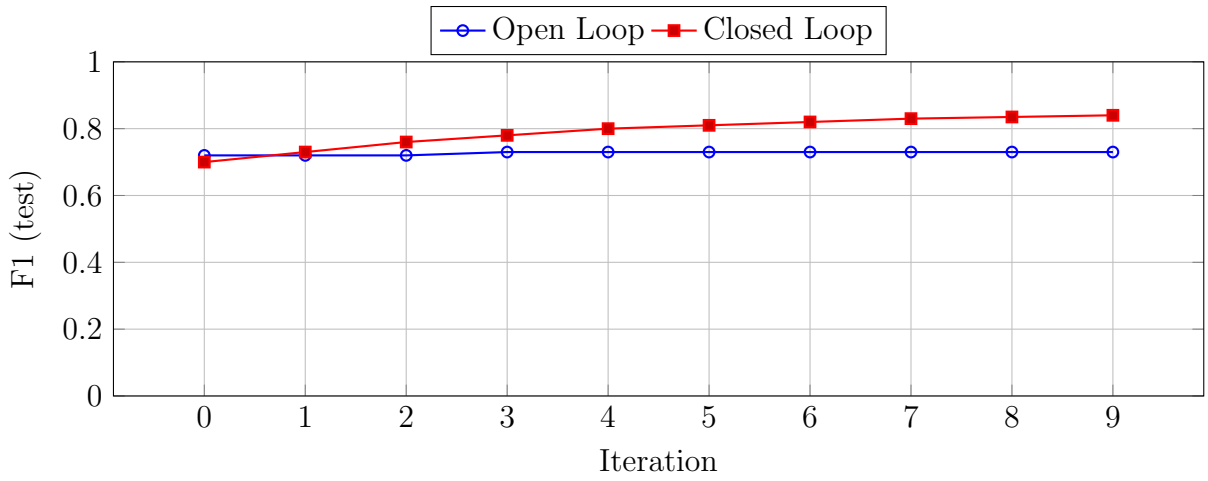


Figure 6.8: F1 across 10 iterations

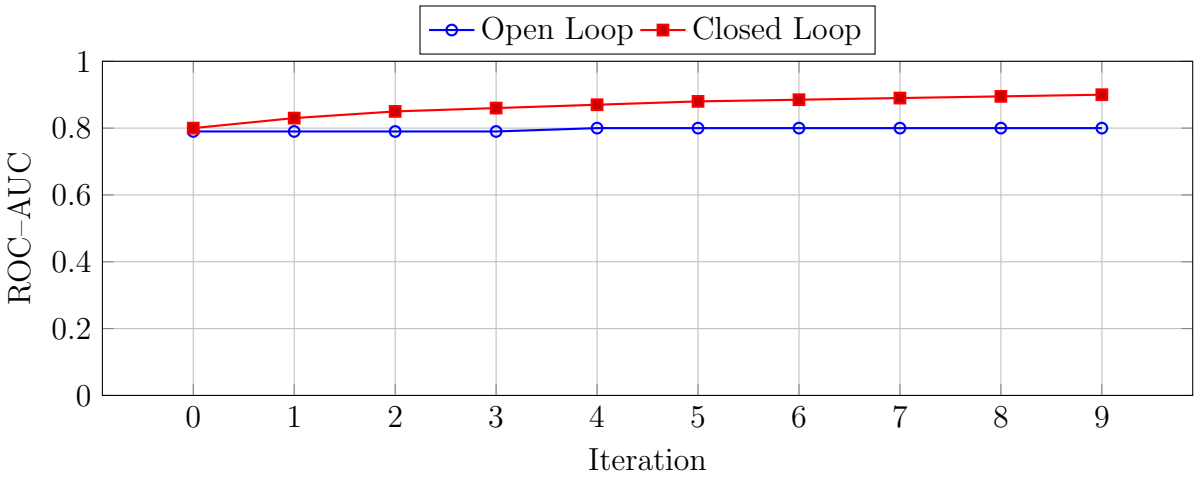


Figure 6.9: ROC–AUC across 10 iterations

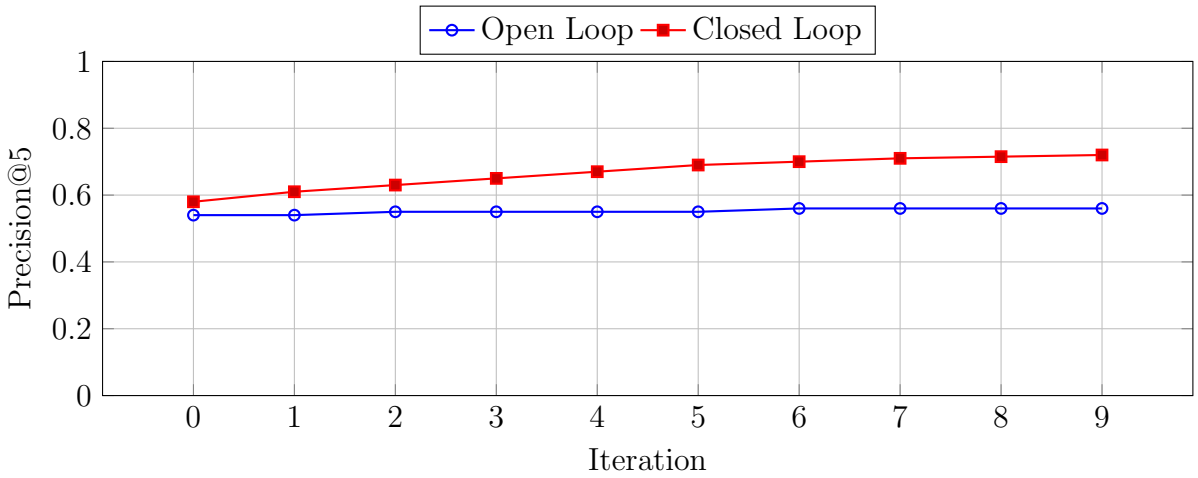


Figure 6.10: Precision@5 across 10 iterations

Chapter 7

Discussion

The testing processes conducted in Chapter 6 provides the final results and comprehensive outline into the effectiveness of internalized feedback loops in a closed looped architecture for the collection and classification of scam related content in an evolving environment. The detailed analysis of each agent evaluation and the overall review of system architecture comparisons will provide insight into how the system architecture facilitates the implementation of AI agent mechanisms in system operations. Specific AI agents such as the data collection agent will provide insight into the specific results required to draw conclusions for the evaluation of web scraping and crawling techniques to scrape dynamically loaded data on sites to collect social media and communication content.

7.1 Agent Analysis

7.1.1 Modern Web Scraping and Crawling in conjunction with Large Language Models(RQ2., RQ2.a)

The analysis across each AI agent in the scam detection system reveals notable differences in results for the closed loop architecture implementations. The data collection agent metrics in table 6.2 indicate that through the control of system search functions through the orchestrator agent an overall increase in pages processed for a piece of content was achieved. The Discovery success rate in Table 6.2 displays the difference in number of sites successfully proceed for an item and tells us that when given the opportunity the data collection can collect material from more sources then originally provided. The implementation of the data collection agent also provides clarity to how modern web scraping tools and techniques can be used in conjunction with large language models to collect data across multiple domains. Figure 4.4 provides the outline for the implementation of the text-embedding-3-small model for individual content processing to retrieve items to assist in the classification agent process.

Higher discovery rates are evident in the comparison from table 6.2, giving a key piece of evidence to support the theory that LLM powered assistance will lead to an increase in data collected for processing. When considered with the additional information that in a closed loop system, the data collection agents experiences a higher overall average throughput compared to the open loop implementations. Higher processing rates included the scam detection and classification system implemented is able to continuously scrape for potential scam data without any limitations or crawl restrictions due to failing to load page content. It can also be concluded that the Unique URL ratio metric gives evidence to suggest that with the assistance of the Orchestrator agents LLM assessment the data collection agent was able to successfully process more unique sources.

Impact of AI Interactions(RQ1.b)

The systems handling of multiple data sources and the additional manual processing of page content through the collection of libraries demonstrated the agents capability to handle data sources of all type as well as those with heavy dynamic loading content. The Process success rate (97%) concluded during the testing process demonstrates the data collection agents ability, but its final success must also consider the initial success rate (88%) as both share the same implementation outside of search adjustments through iterations. This produces a shifting dataset through AI interactions which can be considered to skew results in favor of the closed loop agent but also highlights a limitation in the baseline implementations.

7.1.2 Scam Classification (RQ3.)

The classification agent in the scam detection and classification system is responsible for providing all insights into weather LLMs can be used for the classification of complex datasets. The findings demonstrate both the capabilities and limitations of the current implementations for the scam classification agent through the testing process. The LLM analysis of item embeddings, scam content, and additional contextual information exceeds the target metrics with a satisfactory failure rate to include wrong classifications. Hence we are given the Precision and recall results from testing this process, providing evidence to support that the classification task successfully achieved its goal while still being able to adapt to an evolving environment. The recall (0.98) also indicates that through the processing of sites from the given initial test configuration little errors occurred leading to the inability to extract an item embedding from the page.

This evidence addressed RQ3 with context to the direct utilization of large language models for the classification process. Similarly the identification process is outlined through the results from the data collection agent yet both still implement endpoints for external adjustments from another LLM powered AI agent. The orchestrator agent in the anti scam system adjusts both the search function for the data collection agent and the RAG example pool for the classification agent, leading to effects on their outputs. When

considering Figure 6.3 for the results of altering the pool and number of exemplars to be selected from we can see the relevance to the classification content reduces. Although the closed loop still demonstrates a decreased rate these results can be limited through the orchestrator agent and eliminating the reduction to set boundaries.

7.1.3 Scam Trend Analysis (RQ3.a)

In answering research question 3.a both the individual impact of results on the system and individual scam trend analysis agent outputs must be analysed. While conducting the individual agent testing process, the number of calls made due to the analysis of a scam trend report to the orchestrator agent depicted the impact a result had on the operations of the system. The impact they had was only considered in the difference it cause rather than if that change was positive or negative. Figure 6.4 demonstrates that over time the number of calls steadily impacted the system operations, this in conjunction with the results from the number of calls made in figure 6.5 and positive trend in metric data from the system overall tell us the change in trajectory is positive. Considering these factors it can be concluded that the system successfully analysis scam trend data to cause an impact in the system operations.

Yet despite this significant flaws can be seen in the individual scam trend analysis metrics in regards to successfully identifying trends in the data. The system is able to achieve clustering to a specific point but not what is expected of a system that is able to achieve real time trend analysis for its environment. Over time these trends will be identified and shifted from one group to another maintaining overall metric achievements but not truly displaying the ability to adapt in real-time. The orchestrator and closed loop system architecture results in Chapter 6 must be evaluated alongside the scam trend analysis results to provide context to the impact that the AI implementation has on the system through the final output of the orchestrator agent. In each of the 10 iteration tests for the system the closed loop architecture outperforms the open loop architecture indicating that over a period of time, a result from the trend analysis agent has caused a reaction from the orchestrator agent and in turn impacted the overall system operations leading to a positive growth in metric analytics.

7.2 System Analysis

7.2.1 Closed Loop Multi-Agent Architecture(RQ1,RQ4)

The individual agent analysis serves as the the initial analysis required to complete the overall evaluation of how AI driven agents and web scraping techniques can be combined to effectively detect and track evolving scam activity on social media and communication platforms. Additionally each individual testing procedure is an analysis process conducted to provide an answer to research question 4. The data collection agent demonstrated an overall improvement in data collection and number of unique sources processed in the

closed loop architecture. Leading from the sources of data the scam classification agent in the closed loop architecture systems classifications rates achieved the target metrics, validating that the sourced material was successfully analysed. Comparing the metrics provided in table 6.3 indicates that the increase is prevalent in all tests and also leads to grated success in the trend analysis agent further supported through the metrics in table 6.4. Final comparisons of the architectural influence on the system results in both the analysis regarding the achievement of the orchestrator agent and weather a closed loop knowledge sharing multi agent architecture with an orchestrator agent is able to outperform an open loop feed forward architecture.

The validation process implemented in this research covers the overall evaluation of architectural types in chapter 5 through four metrics displayed in a table comparison. The scam detection and collection systems results indicate that the closed loop architecture achieved higher results in comparison to the open loop architecture, with the only metric indicating an instance were their is no performance benefit in in the initial iterations of testing the system with the same configuration settings. This is due to the fact that at the start of each testing phase the architectures use the same initial configuration settings following the testing methodology outlined in this thesis.

The final system results from the testing process each indicate a different aspect that contribute to the overall successful evaluation of the system in its entirety. Evaluating the quality of data, precision and ability for the system to successfully complete its task at a given point are the foundational components for how the system demonstrates its ability. The comparison provided in the results shows that a change indicates that the system made alterations based on the assessment from an LLM evaluation on scam trend data. This feedback loop in the closed loop architecture and the improvement in individual agent results per implementation is how the system achieves its goal in out performing the open loop architecture.

7.2.2 Research Limitations

The testing procedural approach implemented in the project is limited to the currently developed dataset and preliminary real world testing for an evolving environment. While the testing methodology still provides valuable insight and metrics for evaluation real world testing to achieve a successful and accurate metric for the ability to adapt in an evolving environment in real time would provide the required data to evaluate the system further.

Testing the closed loop multi-agent architecture also required the creation of baseline evaluation metrics for the evaluation process due to the innovative approach and limited research available regarding similar systems and overall performance and functionality metrics.

Chapter 8

Conclusion

This thesis supports the exploration, research, design and development of an scam detection and classification system to detect and track evolving scam activity on social media and communication platforms. The both a design centric research methodology and systematic development process, the study demonstrated the ability to leverage web scraping and AI agents for scam detection on social media and communication platforms.

The testing methodology and process reveals the systems capabilities to effectively adapt to a changing environment in real time in order to collect and classify potential scam data. The system displays significant strengths in the data collection and classification process with the interaction of internalized self learning loops to assist in AI agent operations. Evaluation further validates that the system is capable of achieving success in real work testing through the success in the final iterations of the testing process. Most notable we see an overall increase in the successful processing of unique data sources and continuous classification of evolving datasets with sustained accuracy.

The testing process and evaluation of results also identify areas requiring improvement and further development in order to improve system operations and limit the impact of negative influences such as that through high exemplar pools for the RAG implementation in the classification agent.

Overall the thesis aims to research and contribute to the software engineering industry and understanding of the ability to leverage web scraping and AI agents for scam detection. Each chapter aims to successfully provide the necessary information to support the findings in this paper in regards to the overall problem as well as exploring an innovative solution to evaluating a closed loop knowledge sharing multi agent architectures advantages for real time scam detection and classification on social media and communication platforms.

Chapter 9

Future Work

Future work for the proposed closed loop multi agent system builds upon the findings from the testing process regarding the limitations of the system and AI agent functionalities. The primary technical focus identified from the testing procedure is the implementation of a LLM in the trend analysis agent for the labeling of cluster data and providing an assessment of the resulting outputs. Improvements in this area would implement different data analysis models and procedures to identify a model that is capable of achieving greater results to then be given to a LLM to summarize and provide an assessment.

Further evaluating the system for changes requires continuous testing and real work testing iterations in order to outline the limitations of the system through operation in both an evolving real world environment and in real-time with changing scam classification and trend data. Additionally large scale implementations of the data collection agent following the architectures ability to modularize agent operations will provide valuable information regarding the development limitations in regards to fault-tolerance and heavier system load testing.

Chapter 10

Abbreviations

LLM	Large Language Model
RAG	Retrieval-Augmented Generation
AI	Artificial Intelligent
OSINT	Open Source Intelligence
GASA	Global Anti-Scam Alliance
ACCC	Australian Competition and Consumer Commission
FTC	Federal Trade Commission
NASC	National Anti-Scam Center
OSINT	Open Source Intelligence
URL	Uniform Resource Locator
AI-Agents	Artificial Intelligent Agents
FIPA	Foundation for Intelligent Physical Agent
ACL	Agent Communication Language
NLP	Natural Language Processing
CLMA	Closed Loop Multi Agent
FFP	Feed Forward Agent Pipeline
AI	Artificial Intelligent

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Appendix A

Example of Iterative Prompt Refinement

```
classifier_prompt = f"""You are an AI assistant specializing in scam detection in social media content.
Your task is to classify the following social media post as either legitimate, or one of the scam types listed below.

Scam categories:
- legitimate
- phishing link
- investment scam
- impersonation scam
- fake shopping scam

Review your post and determine the single best-fitting category.
Provide only the category name as an output. nothing else.

Post: {test_data}
"""
```

Figure A.1: Initial Prompt

```
def make_classifier_prompt(test_data):
    return f"""You are an AI assistant specializing in scam detection in social media content.
Your task is to classify the following social media post as either legitimate, or one of the scam types listed below.

Scam categories:
- legitimate
- phishing link
- investment scam
- impersonation scam
- fake shopping scam

Review your post and determine the single best-fitting category.
Provide only the category name as an output. nothing else.

Post: {test_data}
"""
```

Figure A.2: Develop and Refine for Implementation [Second Iteration]

```
def make_classifier_prompt(test_data):
    return f"""You are an AI assistant specializing in scam detection in social media content.
Your task is to classify the following social media post as either legitimate, or one of the scam types listed below.

Scam categories:
- legitimate
- phishing link
- investment scam
- impersonation scam
- fake shopping scam

Examples:
1. "Your account has been suspended, verify your details at http://comnbank.com" → phishing link
2. "I just doubled my money in this crypto investmnet, DM me to join this Group!" → investment scam
3. "ANZ is giving away a free $500 gift card, click to claim!" → impersonation scam
4. "Nike sneakers at 90% off, limited time only at http://cheapnike.com" → fake shopping scam

Review the post and determine the single best-fitting category.
Provide only the category name as an output, nothing else.

Post: {test_data}
"""
```

Figure A.3: Refine Prompt by Adding Examples [Third Iteration]

Appendix B

Type of Social Media Site Scraped (Limited to Pages and Groups in Prototype)

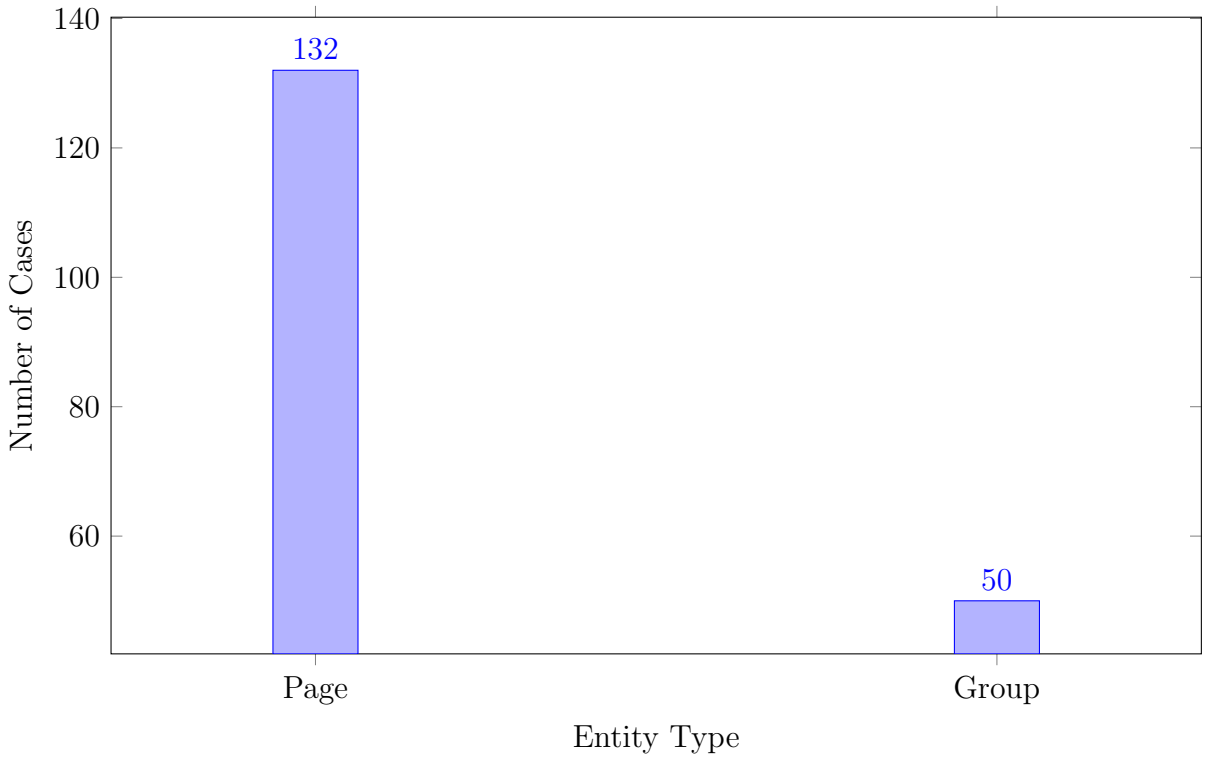


Figure B.1: Distribution of scam cases by Facebook entity type

Appendix C

Preliminary Results

This section presents the preliminary results for an initial prototype of my proposed solution. The data for the initial system was collected from a restricted implementation of the Web Scraper Agent component limited to the collection of data using the OSINT gathering technique. The second restriction was the filtering of data that did not achieve a scam score of six or higher and will be used as a means for initial training data for the development process to be conducted in Thesis B.

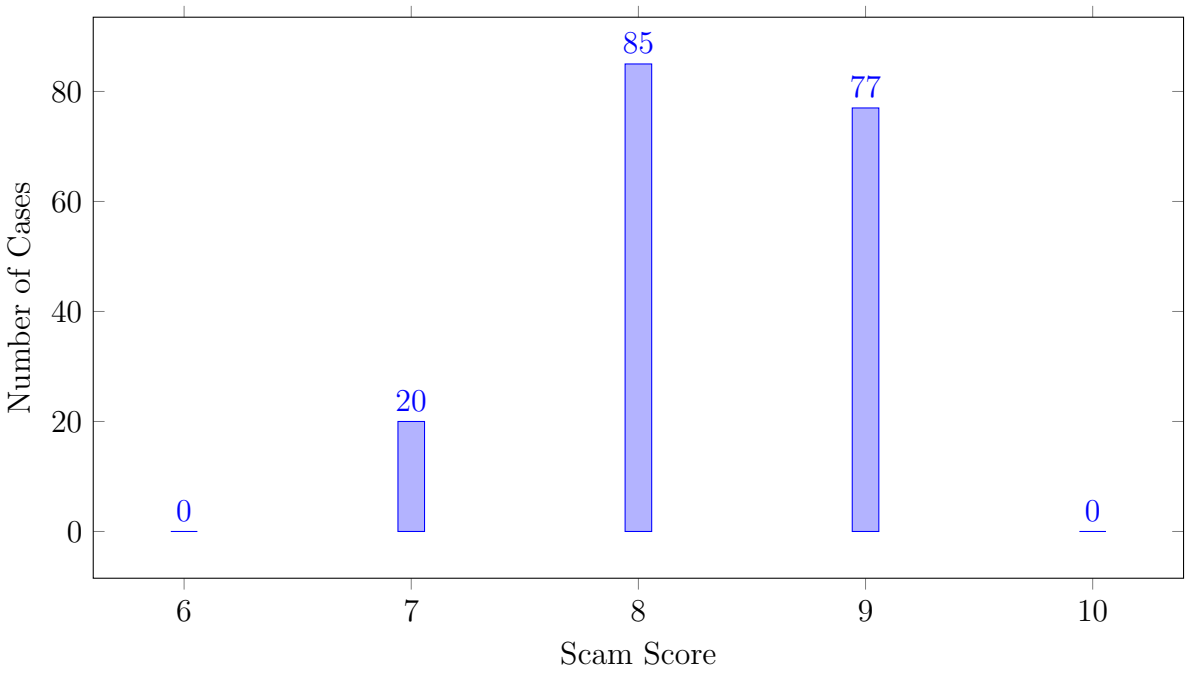


Figure C.1: Distribution of scam scores

The Preliminary results were classified and provided a confidence score using OpenAI's 'gpt-4o'; the advanced reasoning model was selected for its outstanding ability to handle

complex tasks and ability to filter out data considered noise with chained prompting. Figure 5.1 displays the collective results for the number of cases that achieved a scam score and while filtering was implemented to ensure sufficient training examples for models to be implemented in Thesis B, in a total of 244 entries 74% of those were classified as highly potential scams. Additionally in Figure 5.2 an overwhelming amount were classified as Investment Scams or Impersonation Scams, both of which could currently be considered as misrepresentation of data as the little literature regarding the use of LLM's for scam classification identified low levels of accuracy for the task when compared to other Machine learning or Deep Learning algorithms.

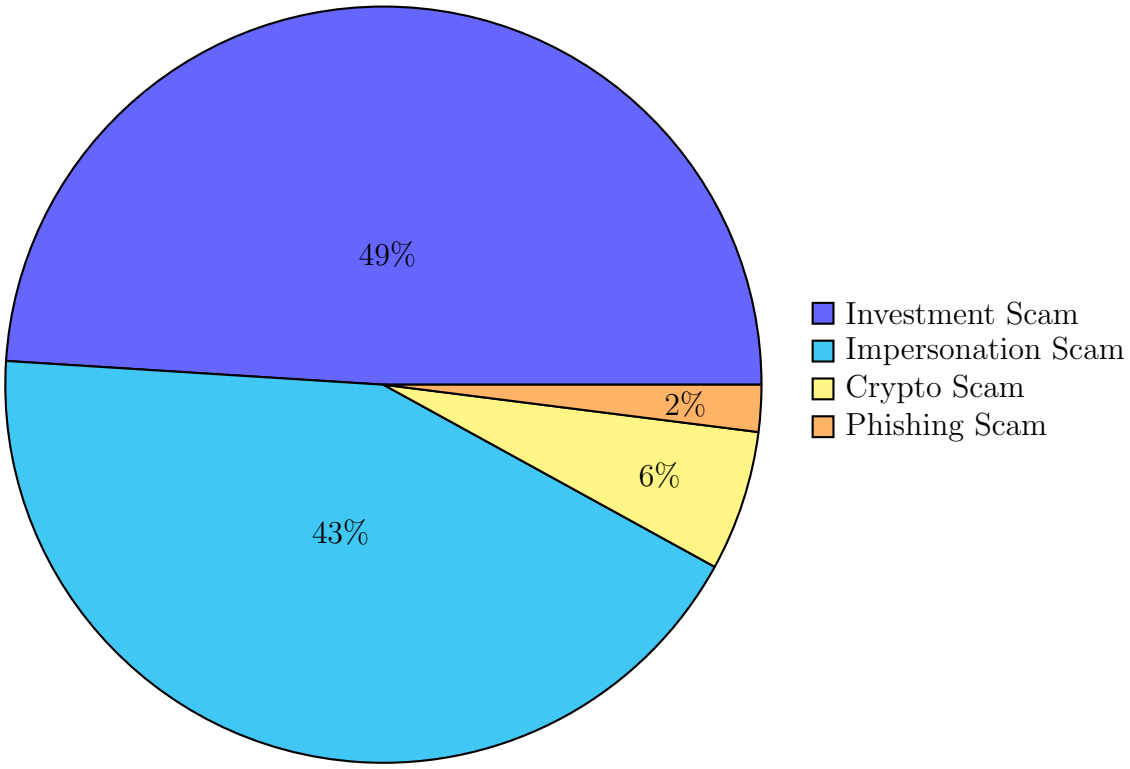


Figure C.2: Distribution of scam types following preliminary prompt examples

Although this may be the case, manual review of 40 records concluded that those identified as 'Crypto Scams' and 80% of those manually reviewed as investment scams where successfully classified.

Appendix D

Data Collector Extraction and Embedding Processing

```python
def parse_extraction_response(self, response: str, url: str) -> URLContentAnalysis:
    try:
        parsed = json.loads(response)
        entities_data = parsed.get('entities', {})
        entities = Entities(
            people=entities_data.get('people', []) if isinstance(entities_data.get('people'), list) else [],
            organizations=entities_data.get('organizations', []) if isinstance(entities_data.get('organizations'), list) else [],
            locations=entities_data.get('locations', []) if isinstance(entities_data.get('locations'), list) else [],
            products=entities_data.get('products', []) if isinstance(entities_data.get('products'), list) else []
        )

        return URLContentAnalysis(
            url=url,
            title=str(parsed.get('title', 'Unknown')),
            main_content=str(parsed.get('mainContent', '')),
            key_topics=parsed.get('keyTopics', []) if isinstance(parsed.get('keyTopics'), list) else [],
            content_type=self.validate_content_type(parsed.get('contentType')),
            entities=entities,
            sentiment=self.validate_sentiment(parsed.get('sentiment')),
            trustworthiness=max(0, min(1, float(parsed.get('trustworthiness', 0.5)))),
            reasoning=str(parsed.get('reasoning', 'No reasoning provided'))
        )
    except Exception as error:
        print(f'Error parsing extraction response: {error}')
        return self.get_default_analysis(url)

async def generate_embeddings(self, text: str) -> List[float]:
    try:
        response = self.openai.embeddings.create(
            model="text-embedding-3-small",
            input=text[:8000]
        )
        return response.data[0].embedding
    except Exception as error:
        print(f'Error generating embeddings: {error}')
        return []
```

Figure D.1: Data Collection Agents Implemented Function for Feature Extraction and Embedding Processing